

Asymptotically Optimal Mixed-State Cloning

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Abstract

Unknown quantum states can only be cloned approximately. Although the optimal pure-state cloner has long been known, the mixed-state problem remains challenging. We give an asymptotically optimal mixed-state cloning protocol, evaluated by global fidelity. Let $d \geq 2$ and $m_n/n \rightarrow \gamma > 1$. We determine the asymptotically optimal global $n \rightarrow m_n$ cloning root fidelity for full-rank d -dimensional states with a non-degenerate spectrum. For a known spectrum $p_1 > \dots > p_d > 0$, the minimax fidelity on its unitary orbit is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}}, \quad q_{ij} = \frac{p_j}{p_i}.$$

If the spectrum is unknown, a single spectrum-independent cloner guarantees the pointwise benchmark

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} F_{\text{orb}}(\gamma, p).$$

On every regular compact spectral set K , the global minimax value is $\inf_{p \in K} F_{\text{univ}}(\gamma, p)$. The cloner is constructed by combining Schur–Weyl label processing with a Cartan-sector covariant channel. For the converse, two-way local asymptotic normality transfers any cloner to classical Gaussian dilation and quantum Gaussian amplification problems, whose optimal bounds give the matching converses.

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1 Introduction

The no-cloning theorem rules out perfect copying of an arbitrary unknown quantum state [WZ82, Die82]. Approximate cloning asks instead how well a quantum channel can transform n identical inputs into an output resembling m ideal copies. For pure states, early universal qubit machines were followed by exact solutions for the standard global and single-clone criteria [BH96, GM97, BDE+98]: Werner solved the global universal problem in arbitrary dimension [Wer98], while Keyl–Werner established optimality when individual clones are tested [KW99]. State-dependent global cloning was developed in [BDE+98, CB99]; see also the reviews [SIGA05, FWJ+14].

The choice of figure of merit is essential. At the level of individual clones, the many-output limit is closely tied to state estimation [MP95, BEM98, BA06, CD06]. When all outputs are examined jointly, the relation is subtler: global asymptotic cloning has been analyzed for broad pure-state families, and an optimal measure-and-prepare output need not consist of independent copies of the estimated state [CY14]. The dependence of the optimum on a joint, marginal, or process-level criterion is already visible for coherent-state and phase-covariant families [CKNWW05, KC22]. Here we use root fidelity with the entire product target $\rho^{\otimes m}$. This global criterion is stronger than reproducing one-site marginals and is therefore distinct from broadcasting, which permits correlations among the outputs [BCFJS96].

Mixed states do not evade the exact no-cloning obstruction: the general no-broadcasting theorem rules out a single channel that clones every member of a noncommuting family of density matrices, with an entropic proof given by Kalev–Hen [BCFJS96, KH08]. Approximate mixed-state cloning poses an additional structural difficulty: both the eigenvalues and the eigenbasis may be unknown, and the two kinds of information behave differently under repeated sampling. In contrast to the pure-state case, earlier mixed-state results either did not identify an optimal cloner with a matching value or applied only to special state families or different criteria. Thus no sharp general solution was known under the global product-fidelity criterion. Work directly on mixed-state cloning derived global-fidelity, relative-error, and state-dependent bounds for prescribed finite ensembles [Ras03a, Ras03b, Ras03c]; constructed cloners for symmetric-subspace inputs and identical mixed

qubits under one-site shrinking-factor criteria [Fan03, FLS07]; studied a restricted mixed-qubit construction under an averaged Hilbert–Schmidt criterion [Kaz10]; and established general entropic limitations on simultaneous approximate cloning and broadcasting of mixed states [LW17]. Closely related work solved optimal mixed-qubit purification and a local-fidelity cloning/estimation problem, rather than the joint product-fidelity task considered here [CEM99]. Multi-input mixed-state work has also focused on broadcasting and superbroadcasting, which optimize marginals rather than the joint product target [DMP05, BDMP05, BDMP06, DF07]. More recently, the random-purification channel of Tang–Wright–Zhandry was combined with pure-state cloning to give a general all-output guarantee, rather than an exact optimum, for rank-constrained mixed states [TWZ25, LTHC26].

In general, mix-state cloning problem admits two models. In the known-spectrum problem, the eigenvalues are supplied and the state ranges over one unitary orbit; cloning must reproduce only the unknown eigenbasis. In the unknown-spectrum problem, a single spectrum-independent sequence of channels must handle both the eigenvalues and the eigenbasis, uniformly over compact subsets of the strict spectral chamber. The second problem is not merely the first preceded by spectrum estimation: the measured spectrum retains fluctuations of order $n^{-1/2}$, which must themselves be converted from the n -copy to the m_n -copy scale.

Here we resolve the mix-state cloning problem in the proportional-output asymptotic regime. For every fixed dimension and $m_n/n \rightarrow \gamma > 1$, we determine the optimal global cloning fidelity and construct asymptotically optimal cloners both on each full-rank, nondegenerate unitary orbit and, when the spectrum is unknown, uniformly over regular compact spectral sets.

Two complementary decompositions drive the solution. Schur–Weyl duality separates the spectral information, carried by the Young label, from the eigenbasis information, carried by the conditional irreducible state [KW01]. Our cloners process the label classically and apply a multiplicity-one Cartan-sector channel to the irreducible state [PRV67, FH91]. Two-way local asymptotic normality (LAN) gives the corresponding asymptotic separation into a classical Gaussian shift and displaced thermal oscillator modes [GJ07, GK06, KG09]. It transfers any putative improvement to a Gaussian dilation or amplification problem. Quantum amplifier noise limits [Cav82] and the least-noise mixed-Gaussian cloner [GM06] identify the quantum benchmark; a unified flat-prior fidelity argument gives the classical and quantum converses used here.

1.1 Results

Fix positive integers m_n such that

$$\gamma_n := \frac{m_n}{n} \longrightarrow \gamma > 1, \quad r_d := \binom{d}{2}.$$

In particular, $m_n > n$ for all sufficiently large n . For the finitely many indices with $m_n \leq n$, the two cloners constructed below are both defined by the marginal channel

$$\mathcal{C}_n^p(X) = \mathcal{C}_n^{\text{univ}}(X) := \text{Tr}_{m_n+1, \dots, n}(X),$$

with the empty partial trace interpreted as the identity when $m_n = n$. This channel is spectrum independent, $\text{U}(d)$ -covariant, and sends $\rho^{\otimes n}$ exactly to $\rho^{\otimes m_n}$. Thus all nontrivial construction formulas and asymptotic proofs below concern $m_n > n$. Write $\mathbf{A}_d := \{x \in \mathbb{R}^d : \sum_i x_i = 1\}$. Let

$$\Delta_d^\circ := \{p \in \mathbf{A}_d : p_1 > \dots > p_d > 0\}, \quad \rho_p = \text{diag}(p_1, \dots, p_d), \quad \rho_{p,g} = g\rho_p g^*.$$

For positive trace-class operators A, C , not necessarily normalized, we write

$$F(A, C) = \|\sqrt{A}\sqrt{C}\|_1.$$

For states this is the unsquared Uhlmann fidelity. For nonnegative functions or measures, the same notation denotes their Hellinger affinity; in particular, $F(P, Q) = \sum_x \sqrt{P(x)Q(x)}$ on a countable space. The target in every statement below is the entire product state $\rho_{p,g}^{\otimes m_n}$. For $p \in \Delta_d^\circ$, define

$$q_{ij} = \frac{p_j}{p_i}, \quad f_\gamma(q) = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q},$$

and

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_\gamma(q_{ij}), \quad F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2}, \quad F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p). \quad (1)$$

The factor F_{orb} measures the cost of amplifying the unknown eigenbasis, whereas F_{cl} measures the additional cost of dilating the unknown spectral fluctuation.

Known spectrum Fixing p isolates the genuinely noncommutative part of the problem. The target distribution of Young labels is then known and can be sampled exactly at size m_n ; only the conditional representation sectors, which encode the eigenbasis, need to be cloned.

Theorem 1.1 (Known-spectrum optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = F_{\text{orb}}(\gamma, p), \quad (2)$$

where the supremum is over all quantum channels from n to m_n qudits. The explicit sequence of channels $\mathcal{C}_n^p$ defined in [Section 2.3](#) asymptotically attains the displayed value.

Thus every pair $i < j$ contributes one displaced-thermal amplification factor $f_\gamma(p_j/p_i)$. The Schur–Cartan cloner attains their product, while fixed-spectrum LAN shows that any better finite-dimensional cloner would violate the Gaussian amplification optimum proved here.

Unknown spectrum If p is not supplied, the target Young-label distribution can no longer be sampled directly. The measured input label determines the spectrum only at the $n^{-1/2}$ scale, and converting n copies into $m_n \sim \gamma n$ copies requires dilating this classical fluctuation. This produces the extra, spectrum-independent factor $F_{\text{cl}}(\gamma, d)$.

Theorem 1.2 (Unknown-spectrum achievability and compact-set optimum). *There exists a sequence of spectrum-independent, $\mathcal{U}(d)$ -covariant channels $\mathcal{C}_n^{\text{univ}}$ with the following properties.*

(a) *For every compact $K \Subset \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \inf_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \mathcal{U}(d)}} \left[F(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) - F_{\text{univ}}(\gamma, p) \right] \geq 0. \quad (3)$$

(b) *Let $K \Subset \Delta_d^\circ$ be nonempty and satisfy $K = \overline{\text{int}_{\mathbf{A}_d} K}^{\mathbf{A}_d}$. Then*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \mathcal{U}(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \inf_{p \in K} F_{\text{univ}}(\gamma, p), \quad (4)$$

where the supremum is over all quantum channels from n to m_n qudits. The same sequence $\mathcal{C}_n^{\text{univ}}$, defined in [Section 2.4](#), asymptotically attains the displayed value.

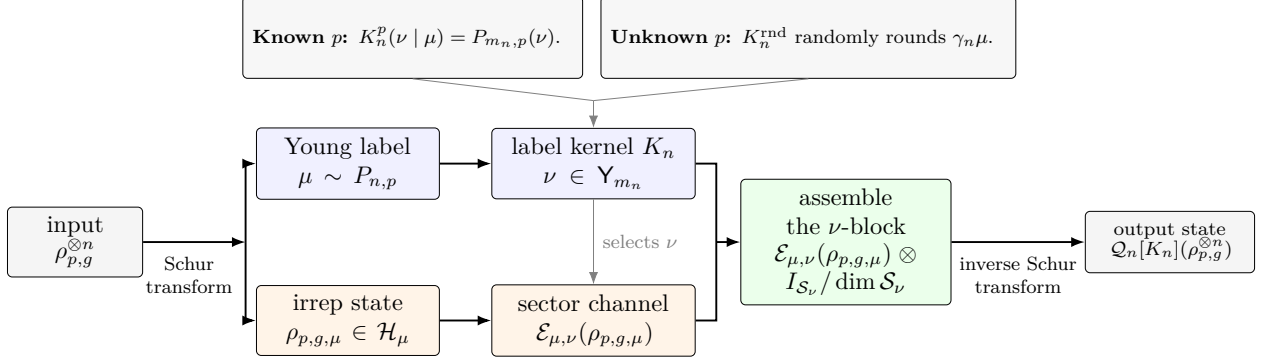


Figure 1: **Cloner Construction.** The Schur transform separates the classical Young label from the conditional irrep state. A Markov kernel selects the output label, while the generic sector channel $\mathcal{E}_{\mu,\nu}$ maps the conditional state into that sector; a maximally mixed Specht factor is then restored. Here $\mathcal{Q}_n[K_n]$ denotes the lifted channel associated with the selected kernel. For known spectrum the target label is sampled independently of μ , whereas for unknown spectrum a randomized dilation is used; these choices produce our cloners for $m_n > n$, namely $\mathcal{C}_n^p = \mathcal{Q}_n[K_n^p]$ and $\mathcal{C}_n^{\text{univ}} = \mathcal{Q}_n[K_n^{\text{rnd}}]$, respectively. On the label pairs carrying asymptotically all probability, the sector fidelity tends to $F_{\text{orb}}(\gamma, p)$, giving the displayed benchmarks.

The common architecture of $\mathcal{C}_n^p$ and $\mathcal{C}_n^{\text{univ}}$ is summarized in Fig. 1. After the Schur transform, the Young label μ carries the spectral information, while the conditional irrep state carries the unknown eigenbasis. A classical kernel chooses an output label ν , and the common covariant sector channel maps the conditional state into the ν -sector before the m_n -system output is reconstructed. For known p , the label kernel samples the exact target law $P_{m_n,p}$ independently of μ ; the spectrum-independent kernel instead randomly rounds $\gamma_n \mu$. Their label factors are therefore 1 and F_{cl} , while the common sector factor is F_{orb} , giving the totals F_{orb} and F_{univ} . The details are developed in Section 2.

The minimax meaning of (4) is as follows. The set K may be known when the channel is designed, but the actual spectrum $p \in K$ and eigenbasis g are not. The order of optimization is important: the supremum first allows a channel sequence tailored to K , and the inner infimum then tests that single sequence against every $p \in K$ and $g \in \text{U}(d)$. Since the unknown-spectrum benchmark $F_{\text{univ}}(\gamma, p)$ varies with p , the resulting worst-case scalar is governed by its least favorable value, hence the infimum over p on the right-hand side. No further infimum over g appears there because unitary covariance makes the benchmark constant along the orbit of each fixed spectrum. Thus, even with knowledge of K , no K -adapted sequence of channels can asymptotically beat $\inf_{p \in K} F_{\text{univ}}(\gamma, p)$. The single construction $\mathcal{C}_n^{\text{univ}}$, which does not depend on K , guarantees $F_{\text{univ}}(\gamma, p)$ pointwise, uniformly for $p \in K$, and hence asymptotically attains this worst-case value.

The condition $K = \overline{\text{int}_{\mathbf{A}_d} K}^{\mathbf{A}_d}$ is a clean sufficient regularity assumption for full-dimensional spectral uncertainty. Its interior contains small neighborhoods in all $d - 1$ spectral directions, and the closure condition allows the infimum over interior points in the converse to be replaced, by continuity of F_{univ} , with the infimum over all of K . We do not claim that this condition is necessary, but some thickness assumption cannot be removed from a geometry-independent statement of this form. For a singleton $K = \{p\}$, the channel may be tailored to p , so the problem has the known-spectrum value $F_{\text{orb}}(\gamma, p)$, not $F_{\text{univ}}(\gamma, p)$. More generally, lower-dimensional spectral classes encode partial spectral knowledge and may have different local Gaussian penalties. Thus (4) is the broadest geometry-independent full-dimensional converse proved here; no statement with the same

right-hand side can extend to all compact K .

For example, a natural admissible class keeps the smallest eigenvalue and every adjacent spectral gap uniformly positive. For $0 < \kappa < 2/[d(d+1)]$,¹ set

$$\Delta_{d,\kappa} = \{p \in \mathbf{A}_d : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa \ (i < d)\}.$$

This is a nonempty compact subset of Δ_d° and satisfies $\Delta_{d,\kappa} = \overline{\text{int}_{\mathbf{A}_d} \Delta_{d,\kappa}}^{\mathbf{A}_d}$, so (4) applies with $K = \Delta_{d,\kappa}$. Indeed, choose $\kappa' \in (\kappa, 2/[d(d+1)])$ and set $p_i = 1/d + ((d+1)/2 - i)\kappa'$. Then $p_i - p_{i+1} = \kappa' > \kappa$ and $p_d > 2/[d(d+1)] > \kappa$, so this is a relative-interior point. The set is closed in the probability simplex, hence compact; being convex with nonempty relative interior gives the stated closure property.

1.2 Remarks

Comparison with purify-and-clone. The random-purification channel of Tang–Wright–Zhandry maps n copies of any rank- r state to the uniform mixture of n identical purifications on $\mathbb{C}^d \otimes \mathbb{C}^r$ [TWZ25]. The purify-and-clone strategy of Li–Theil–Harrow–Chuang then applies Werner’s optimal pure-state cloner on the dr -dimensional purification space [Wer98, LTHC26]. Their Eq. (1a) and Theorem S3.1 give the all-output root-fidelity guarantee

$$F_{\text{RP}}^{\text{root}} \geq \left(\frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}} \right)^{1/2} \longrightarrow \gamma^{-(dr-1)/2}.$$

The square root of the binomial ratio is the exact root fidelity of the pure-state cloning step, but after the purifying registers are discarded it is only the displayed lower bound on the mixed-state fidelity. For full-rank inputs, $r = d$, our exact values satisfy

$$F_{\text{orb}}(\gamma, p) > \gamma^{-d(d-1)/4} > \gamma^{-(d^2-1)/2}, \quad F_{\text{univ}}(\gamma, p) > \gamma^{-(d^2-1)/4} > \gamma^{-(d^2-1)/2},$$

so, at each fixed full-rank nondegenerate spectrum in our regime, both give strictly stronger proved guarantees.

The comparison can be strengthened from guarantees to the actual purify-and-clone fidelity. We explicitly show this for qubits in Appendix A: Proposition A.2 proves that the limsup of its root fidelity is strictly below $F_{\text{orb}}(\gamma, p)$ for every full-rank nondegenerate qubit spectrum. At $\gamma = 2$ and $p = (2/3, 1/3)$, this limsup is at most 0.8604, whereas the root fidelities of our spectrum-independent and known-spectrum cloners have limiting lower bounds, respectively,

$$F_{\text{univ}}(2, p) \approx 0.8856, \quad F_{\text{orb}}(2, p) \approx 0.9121.$$

Thus both of our cloners are strictly superior to the actual purify-and-clone protocol in this example.

The proved comparison with the actual purify-and-clone protocol is therefore the qubit result just stated. Appendix A also gives a modewise sketch suggesting that the same argument extends to arbitrary dimension.

¹At $\kappa = 2/[d(d+1)]$, the identity $1 = dp_d + \sum_{i=1}^{d-1} i(p_i - p_{i+1}) \geq \frac{\kappa d(d+1)}{2}$ forces a unique spectrum, so the class is a singleton and its optimum is F_{orb} , not F_{univ} . The compact-set theorem is not asserted uniformly as the spectrum approaches an eigenvalue collision or the boundary of the probability simplex.

Dependence on spectral mixedness The product formula makes precise one sense in which a more mixed state is easier to clone. For fixed $\gamma > 1$,

$$f'_\gamma(q) = \frac{(\gamma - 1)^2}{2\sqrt{q(\gamma - 1 + q)}(\sqrt{\gamma(\gamma - 1 + q)} + \sqrt{q})^2} > 0, \quad 0 < q < 1.$$

Consequently, if $p, p' \in \Delta_d^\circ$ satisfy

$$\frac{p'_j}{p'_i} \geq \frac{p_j}{p_i} \quad \text{for every } i < j, \quad (5)$$

then

$$F_{\text{orb}}(\gamma, p') \geq F_{\text{orb}}(\gamma, p), \quad F_{\text{univ}}(\gamma, p') \geq F_{\text{univ}}(\gamma, p),$$

with strict inequality if any ratio is strictly larger. Equivalently, every logarithmic spectral gap has contracted. This ratio-wise flattening implies that p' is majorized by p , but is stronger than majorization. For $d \geq 3$, majorization alone need not increase the cloning fidelity; for $d = 2$, the single eigenvalue ratio does give majorization monotonicity.

Limitations. Our results concern the proportional-output asymptotic regime and assume a full-rank, nondegenerate spectrum. We do not treat degenerate spectra or rank-deficient states, for which the LAN model and stabilizer structure change. We also do not determine the exact optimal fidelity for finite n . It remains unclear whether the cloners constructed here are already optimal for the corresponding finite-copy problems; we conjecture that they are.

Conjecture: degenerate and rank-deficient spectra We conjecture the following formula for the optimal cloning fidelity when the spectrum can be degenerate or rank deficient. Suppose that p has distinct eigenvalues $\lambda_1 > \dots > \lambda_s \geq 0$, with respective multiplicities $r_1, \dots, r_s$, so that $\sum_a r_a = d$ and $\sum_a r_a \lambda_a = 1$. Set

$$c_\gamma = \frac{2\sqrt{\gamma}}{1 + \gamma}, \quad F_{\text{orb}}(\gamma, p) = \prod_{a < b} f_\gamma\left(\frac{\lambda_b}{\lambda_a}\right)^{r_a r_b}.$$

We conjecture that this $F_{\text{orb}}(\gamma, p)$ is the optimal known-spectrum orbital cloning fidelity. Only pairs of distinct eigenspaces contribute: rotations within a degenerate eigenspace belong to the stabilizer of ρ_p and hence give no oscillator mode. If $\lambda_s = 0$, the zero eigenspace is included as the last block, and each positive-zero pair contributes $f_\gamma(0) = \gamma^{-1/2}$.

For the unknown-spectrum problem restricted to a fixed-rank stratum, suppose that $\lambda_s = 0$, and write $r_0 = r_s$ and $r = d - r_0$. We conjecture that the local minimax fidelity within the rank- r manifold is

$$F_{\text{univ}}^{(r)}(\gamma, p) = c_\gamma^{(\sum_{a=1}^{s-1} r_a^2 - 1)/2} F_{\text{orb}}(\gamma, p).$$

Only the positive-eigenvalue blocks contribute to the classical exponent; the support-kernel directions are pure Gaussian modes and are already included in F_{orb} through the factors $f_\gamma(0)$. This decomposition is suggested by fixed-rank low-rank LAN [LN24]. In particular, if the positive eigenvalues are nondegenerate, then

$$F_{\text{univ}}^{(r)}(\gamma, p) = c_\gamma^{(r-1)/2} \gamma^{-r(d-r)/2} \prod_{1 \leq i < j \leq r} f_\gamma\left(\frac{\lambda_j}{\lambda_i}\right).$$

For the full unknown-state local problem, assume instead that $\lambda_s > 0$. Allowing arbitrary $n^{-1/2}$ -perturbations that may split the degeneracies, we conjecture the value

$$F_{\text{univ}}(\gamma, p) = c_\gamma^{(\sum_a r_a^2 - 1)/2} F_{\text{orb}}(\gamma, p).$$

Here $\sum_a r_a^2 - 1$ is the real dimension of the traceless Hermitian commutant of ρ_p , whose local coordinates should become classical Gaussian variables, while the cross-eigenspace directions give the quantum factors above. In particular, for the maximally mixed known spectrum the product is empty and the orbital fidelity is one.

As a sanity check, the fixed-rank conjecture has the correct pure-state specialization. A rank-one spectrum has one eigenvalue 1 and a zero eigenspace of multiplicity $d - 1$, so

$$F_{\text{univ}}^{(1)}(\gamma, p) = F_{\text{orb}}(\gamma, p) = f_\gamma(0)^{d-1} = \gamma^{-(d-1)/2}.$$

This is exactly the $m_n/n \rightarrow \gamma$ limit of Werner’s optimal pure-state cloning root fidelity [Wer98]:

$$\left(\frac{\binom{n+d-1}{d-1}}{\binom{m_n+d-1}{d-1}} \right)^{1/2} \longrightarrow \gamma^{-(d-1)/2}.$$

1.3 Earlier and related work

Pure-state cloning and state estimation. Universal qubit cloners were constructed in [BH96, GM97, BDE+98]; the arbitrary-dimensional global and single-clone optima were obtained in [Wer98, KW99], while links to finite-copy estimation were developed in [MP95, BEM98]. At the single-clone level, cloning with infinitely many outputs reduces to estimation [BA06, CD06]; global all-output problems have also been studied for broad pure-state families [CB99, CY14, KC22]. These settings differ from our proportional-output mixed-state problem; see [SIGA05, FWJ+14] for reviews.

Finite-dimensional mixed-state cloning. The earlier finite-dimensional literature summarized above provides bounds or constructions for restricted state families and criteria, rather than the exact proportional-output minimax value under joint product fidelity [CEM99, Ras03a, Ras03b, Ras03c, Fan03, FLS07, Kaz10, LW17]. More recently, random purification followed by pure-state cloning gave a universal all-output guarantee for rank-constrained mixed states, but not an exact optimum [TWZ25, LTHC26]; companion work instead amplifies a selected eigenstate [LTHC26b]. See Section 1.2 for the quantitative comparison.

Broadcasting and superbroadcasting. Single-input families are broadcastable precisely when they commute [BCFJS96]; an input-independent-fidelity criterion likewise obstructs universal broadcasting for arbitrary mixed qubits [CC07]. Multiple-input superbroadcasting can nevertheless purify individual mixed-qubit marginals; universal, phase-covariant, and optimal maps are known [DMP05, BDMP05, BDMP06, DF07], as is a continuous-variable version for displaced thermal states [DPS06]. These marginal criteria allow correlated outputs, so they do not determine full tensor-power fidelity. Virtual broadcasting is a related nonphysical extension linking universal cloning and optimal physical approximation, without changing this distinction [PFBC24].

Gaussian cloning and LAN. Continuous-variable cloning grew from coherent-state amplifier–beam-splitter schemes [CIR00, BCI+01]; under joint rather than single-clone fidelity, a non-Gaussian cloner can be optimal [CKNWW05]. Caves established the phase-insensitive amplifier noise limit

[Cav82], and Guță–Matsumoto reduced displaced mixed-Gaussian cloning to least-noise amplification [GM06]. Quantum LAN was formulated statistically in [GJ07] and developed through likelihood-ratio and explicit-channel formulations in [YFG13, GK06, KG09], yielding sharp purification, dilution, and reversal results [BGA11, BGA12]; related Gaussian tasks include amplifying and purifying noisy coherent states [ZC17]. This LAN–amplification lineage underlies our converse. Low-rank qudit estimation has different Gaussian asymptotics [LN24], so boundary models lie outside our full-rank LAN statement.

Other cloning and replication models. Further variants include probabilistic exact cloning [DG98]; economical, phase-covariant, asymmetric, and telecloning schemes [NG98, BCDM00, Cer00, MJPV99]; and cloning of transformations and superreplication [CDP08, CYY13, GCMBC14, DSS15, CYH15, CY16]. A higher-order framework for general channel cloning includes state cloning [SGKS25]; no-signalling yields general cloning bounds [SSD15, HT24], while optimal pure-state cloning and transposition can be complementary channels [BGSQ26]. These models differ from ours in the success rule, family, target object, or replication rate.

1.4 Organization

Section 2 develops the common Schur–Cartan lifting, proves a fidelity bound reducing the cloning problem to a classical label affinity and a conditional sector fidelity, and then defines the exact target-label kernel for known spectrum and the randomized dilation kernel for unknown spectrum. Together with two uniform estimates, these constructions prove the achievability half of Theorem 1.1 and part (a) of Theorem 1.2. Section 3 solves the corresponding Gaussian amplification problems, states the two-way LAN approximation, and transfers the Gaussian bounds back to finite dimension. It proves the converse half of Theorem 1.1 and the compact-set converse in part (b) of Theorem 1.2. The quantitative comparison with random purification is summarized in Section 1.2; the strict comparison is developed in Appendix A. The other appendices supply the fixed-height PBW calculation of the sector fidelity, the Young local limit and randomized-rounding estimate, and the unified flat-prior Gaussian converse.

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Generative AI usage

The original idea for the cloner construction was due to the authors. We then make use of OpenAI’s ChatGPT Sol 5.6 and Anthropic’s Fable 5 to assist with the analysis. The evaluation of the cloning fidelity and the converse argument were benefited from suggestions by AI. The AI tools was conducted to assist with language editing, manuscript organization, and figure preparation. All AI-assisted material was reviewed and revised by the authors, who take full responsibility for the accuracy and content of the manuscript.

Glossary of notation

We collect the notation that recurs across sections. Symbols local to a single proof are defined where they first appear.

Spaces and maps. $\mathbb{C}$ and $\mathbb{R}$ denote the complex and real numbers, $U(d)$ the unitary group, and $\mathcal{T}_1(\mathcal{H})$ the trace-class operators on a Hilbert space $\mathcal{H}$. We write A^* for the adjoint, $\text{Tr } A$ for the trace, $\|A\|_1$ for the trace norm, and $\otimes$ for tensor products. A channel is a completely positive trace-preserving map.

Asymptotic scale. The local dimension d is fixed, n is the number of inputs, m_n the number of outputs, and $\gamma_n = m_n/n \rightarrow \gamma > 1$. The number of pairs $i < j$ is $r_d = \binom{d}{2}$. A capital N denotes a generic sample size.

Spectra and states. $A_d = \{x \in \mathbb{R}^d : \sum_i x_i = 1\}$, and Δ_d° is its strictly ordered, positive spectral chamber. For $p \in \Delta_d^\circ$, $\rho_p = \text{diag}(p)$ and $\rho_{p,g} = g\rho_p g^*$. The notation $K \Subset \Delta_d^\circ$ means that K is a compact subset of this open chamber; relative interiors and closures are taken in A_d .

Fidelity and optimal values. The unadorned $F(A, C)$ denotes root (unsquared) fidelity; for classical measures it denotes Hellinger affinity. With $q_{ij} = p_j/p_i$, the one-mode factor is $f_\gamma(q_{ij})$. The quantities F_{orb} , F_{cl} , and $F_{\text{univ}} = F_{\text{cl}}F_{\text{orb}}$ are respectively the orbital, classical, and universal root fidelities in (1).

Schur–Weyl notation. Y_N is the set of partitions $\lambda \vdash N$ with at most d rows. The corresponding unitary and symmetric-group spaces are $\mathcal{H}_\lambda$ and $\mathcal{S}_\lambda$. The Young-label law under $\rho_p^{\otimes N}$ is $P_{N,p}$, its conditional irreducible state is $\rho_{p,\lambda}$, and $\mathbb{T}_N(p)$ is the typical-label set.

Cloners and label kernels. $\mathcal{M}_n$ denotes a generic competing channel. A Markov kernel $K_n(\nu \mid \mu)$ on Young labels is lifted to the covariant channel $\mathcal{Q}_n[K_n]$. For compatible labels, $\mathcal{C}_{\mu,\nu}$ is the Cartan-sector channel; $\mathcal{E}_{\mu,\nu}$ extends it to all label pairs by a fallback channel. The constructed known- and unknown-spectrum cloners are $\mathcal{C}_n^p$ and $\mathcal{C}_n^{\text{univ}}$, respectively. Thus K denotes a spectral set, while K_n denotes a label kernel.

Local coordinates. $\mathbf{H} = \{h \in \mathbb{R}^d : \sum_i h_i = 0\}$ is the spectral tangent hyperplane. The variables $h \in \mathbf{H}$ and $z \in \mathbb{C}^{r_d}$ describe local spectral and orbital perturbations, respectively; $Y(z)$ is the skew-Hermitian generator of the orbital perturbation, $g_{n,z} = e^{Y(z)/\sqrt{n}}$, Ω_L is a bounded local box, and $v_{n,L}(p)$ is its minimax value.

Gaussian notation. The calligraphic $\mathcal{F}$ denotes one-mode Fock space and $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$, distinct from the roman F for fidelity. The state τ_p is the thermal product in the input and ideal target, while $\tau_p^{(\gamma)}$ is the noisy amplifier output at zero displacement. The states $\Phi_z^{(p)}$ and $\Psi_z^{(p,\gamma)}$ are the displaced input and ideal target built from τ_p . The symbols Θ^{in} and Θ^{tar} include the classical Gaussian factor. The number operator is $\hat{N}$, distinct from the sample size N .

Limits and uniformity. Unless stated otherwise, asymptotics are as $n \rightarrow \infty$ with d and γ fixed. An $o(1)$ term is uniform only over the parameters specified in the accompanying statement.

2 Cloner construction and achievability

Both cloning protocols use the same finite-dimensional architecture. We first recall the Schur–Weyl decomposition in Section 2.1 and lift an arbitrary Markov kernel on Young labels to a covariant quantum channel in Section 2.2. We then choose the exact target-law kernel for known spectrum and the randomized dilation kernel for unknown spectrum in Sections 2.3 and 2.4, respectively, and combine their label estimates with the common sector fidelity in Section 2.5. The two achievability proofs are completed in Section 2.6. We now develop the components summarized in Fig. 1.

2.1 Schur–Weyl decomposition

Let

$$\mathbf{Y}_N = \{\lambda \vdash N : \ell(\lambda) \leq d\}.$$

Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\lambda \in \mathbf{Y}_N} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda,$$

where $\mathcal{H}_\lambda$ is the irreducible $U(d)$ -module of highest weight λ and $\mathcal{S}_\lambda$ is the corresponding Specht module. For $\rho_p = \text{diag}(p_1, \dots, p_d)$,

$$\rho_p^{\otimes N} = \bigoplus_{\lambda \in \mathbf{Y}_N} P_{N,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (6)$$

For $\rho_{p,g} = g\rho_p g^*$, replace $\rho_{p,\lambda}$ by $\rho_{p,g,\lambda} = U_\lambda(g)\rho_{p,\lambda}U_\lambda(g)^*$. The Young law $P_{N,p}$ depends on p , but not on g .

Define

$$\mathbf{T}_N(p) = \left\{ \lambda \in \mathbf{Y}_N : \left\| \frac{\lambda}{N} - p \right\|_\infty \leq N^{-1/3} \right\}.$$

Lemma 2.1 (Uniform Young concentration). *There is a constant $c_d > 0$, depending only on d , such that*

$$\sup_{p \in \Delta_d^\circ} P_{N,p}(\mathbf{T}_N(p)^c) \leq e^{-c_d N^{1/3}}$$

for all sufficiently large N .

Proof. Write $q = \lambda/N$. The Weyl dimension formula gives $\dim \mathcal{H}_\lambda \leq (N+1)^{d(d-1)/2}$. Every weight ξ of $\mathcal{H}_\lambda$ has $\lambda - \xi$ equal to a nonnegative integral combination of positive roots. Since $p_j/p_i < 1$ for $i < j$, this gives $p^\xi \leq p^\lambda$. Hence $s_\lambda(p) \leq \dim \mathcal{H}_\lambda p^\lambda$. The standard bounds $\dim \mathcal{S}_\lambda \leq \binom{N}{\lambda_1, \dots, \lambda_d}$ and $\binom{N}{\lambda_1, \dots, \lambda_d} p^\lambda \leq \exp[-ND(q\|p)]$ give

$$P_{N,p}(\lambda) \leq (N+1)^{d(d-1)/2} \exp[-ND(q\|p)], \quad (7)$$

which is the finite-sample estimate underlying the Keyl–Werner spectrum large-deviation theorem [KW01].

There are at most $(N+1)^{d-1}$ partitions of N with at most d rows. If $\lambda \notin \mathbf{T}_N(p)$, then $\|q - p\|_1 > N^{-1/3}$, and Pinsker’s inequality gives $D(q\|p) \geq \frac{1}{2}N^{-2/3}$. Therefore

$$P_{N,p}(\mathbf{T}_N(p)^c) \leq (N+1)^{d(d-1)/2+d-1} e^{-N^{1/3}/2}.$$

For all sufficiently large N , the polynomial factor is at most $e^{N^{1/3}/4}$. Hence the asserted bound holds with $c_d = 1/4$. $\square$

2.2 Cartan-sector channel and lifting of a label kernel

Let λ, θ be dominant weights, with unit highest-weight vectors v_λ, v_θ . The Cartan component $\mathcal{H}_{\lambda+\theta}$ occurs with multiplicity one in $\mathcal{H}_\lambda \otimes \mathcal{H}_\theta$. Let

$$V_{\lambda,\theta} : \mathcal{H}_{\lambda+\theta} \longrightarrow \mathcal{H}_\lambda \otimes \mathcal{H}_\theta, \quad V_{\lambda,\theta} v_{\lambda+\theta} = v_\lambda \otimes v_\theta, \quad (8)$$

be the normalized intertwining isometry. For $\mu \in \mathbf{Y}_n$ and $\nu \in \mathbf{Y}_m$, write $\nu - \mu \vdash m - n$ when the coordinatewise difference $\nu - \mu$ is a partition of $m - n$. If this holds, put $\omega = \nu - \mu$ and define

$$\mathcal{C}_{\mu,\nu}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^* (X \otimes I_{\mathcal{H}_\omega}) V_{\mu,\omega}. \quad (9)$$

The partial trace $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*)$ commutes with the irreducible $\text{U}(d)$ -action on $\mathcal{H}_\mu$; hence Schur's lemma and a trace comparison give $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*) = \frac{\dim \mathcal{H}_\omega}{\dim \mathcal{H}_\mu} I_{\mathcal{H}_\mu}$. Thus (9) is completely positive by tensoring with $I_{\mathcal{H}_\omega}$, applying $Y \mapsto V_{\mu,\omega}^* Y V_{\mu,\omega}$, and positive rescaling; the identity above makes it trace preserving, while equivariance of $V_{\mu,\omega}$ gives $\text{U}(d)$ -covariance.

For every pair $(\mu, \nu) \in \mathbf{Y}_n \times \mathbf{Y}_m$, define a covariant channel $\mathcal{E}_{\mu,\nu} : \mathcal{T}_1(\mathcal{H}_\mu) \rightarrow \mathcal{T}_1(\mathcal{H}_\nu)$ by

$$\mathcal{E}_{\mu,\nu}(X) = \begin{cases} \mathcal{C}_{\mu,\nu}(X), & \nu - \mu \vdash m - n, \\ \text{Tr}(X) I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu, & \text{otherwise.} \end{cases}$$

Let $K_n : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$, with transition probabilities $K_n(\nu \mid \mu)$, be any Markov kernel. Its Schur–Cartan lifting is the following channel, denoted by $\mathcal{Q}_n[K_n]$: apply the Schur transform, measure μ , discard the input Specht factor, sample ν from $K_n(\cdot \mid \mu)$, apply $\mathcal{E}_{\mu,\nu}$, adjoin $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$, and apply the inverse Schur transform. Each step is completely positive and trace preserving, so their composition is a channel. Because the label measurement and sampling are invariant and every map $\mathcal{E}_{\mu,\nu}$ is covariant, $\mathcal{Q}_n[K_n]$ is $\text{U}(d)$ -covariant.

Proposition 2.2 (Fidelity of a lifted label kernel). *Fix $p \in \Delta_d^\circ$. Let*

$$\Pi_{n,p}(\mu, \nu) = P_{n,p}(\mu) K_n(\nu \mid \mu), \quad Q_{m_n,p}(\nu) = \sum_{\mu} \Pi_{n,p}(\mu, \nu).$$

Suppose $G \subset \mathbf{Y}_n \times \mathbf{Y}_{m_n}$ satisfies $\nu - \mu \vdash m_n - n$ for every $(\mu, \nu) \in G$, and

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) \geq b \quad ((\mu, \nu) \in G)$$

for some $b \in [0, 1]$. Then, for every $g \in \text{U}(d)$,

$$F\left(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) \geq b \left(F(Q_{m_n,p}, P_{m_n,p}) - \sqrt{\Pi_{n,p}(G^c)}\right). \quad (10)$$

Proof. Covariance makes every sector fidelity independent of g . In the output Schur decomposition, the actual state in the ν -block is the mixture of $\mathcal{E}_{\mu,\nu}(\rho_{p,g,\mu})$ with weights $\Pi_{n,p}(\mu, \nu)$, whereas the ideal block is $P_{m_n,p}(\nu) \rho_{p,g,\nu}$. The Specht factors are identical and therefore cancel by multiplicativity of the trace norm under tensor products.

For $Q_{m_n,p}(\nu) > 0$, set

$$r(\nu) = \sum_{\mu: (\mu,\nu) \notin G} \frac{\Pi_{n,p}(\mu, \nu)}{Q_{m_n,p}(\nu)},$$

and set $r(\nu) = 0$ otherwise. The direct-sum identity for fidelity, its scaling under positive scalar multiples, and joint concavity give

$$\begin{aligned} & F\left(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) \\ & \geq b \left(F(Q_{m_n,p}, P_{m_n,p}) - \sum_{\nu} \sqrt{Q_{m_n,p}(\nu) P_{m_n,p}(\nu)} r(\nu)\right). \end{aligned}$$

Since $0 \leq r \leq 1$ and $\sum_{\nu} Q_{m_n,p}(\nu) r(\nu) = \Pi_{n,p}(G^c)$, Cauchy–Schwarz yields

$$\sum_{\nu} \sqrt{Q_{m_n,p}(\nu) P_{m_n,p}(\nu)} r(\nu) \leq \sqrt{\Pi_{n,p}(G^c)} \left(\sum_{\nu} P_{m_n,p}(\nu) r(\nu) \right)^{1/2} \leq \sqrt{\Pi_{n,p}(G^c)}.$$

This proves (10). $\square$

We now specialize the lifted channel by choosing its output-label kernel. The two protocols differ only in this choice: the known-spectrum construction samples the exact target law, whereas the unknown-spectrum construction uses a randomized dilation of the measured input label.

2.3 Known spectrum: exact target-label sampling

Fix $p \in \Delta_d^{\circ}$. For every n such that $m_n > n$, define

$$K_n^p(\nu \mid \mu) = P_{m_n,p}(\nu).$$

Thus ν is independent of μ and has the exact target law. Define the amplification branch by

$$\mathcal{C}_n^p := \mathcal{Q}_n[K_n^p].$$

Pairs with $\nu - \mu \not\vdash m_n - n$ use $\mathcal{E}_{\mu,\nu}(X) = \text{Tr}(X)I_{\mathcal{H}_{\nu}} / \dim \mathcal{H}_{\nu}$ without changing the sampled label ν . Together with the common convention in Section 1.1, this defines the full $U(d)$ -covariant sequence $(\mathcal{C}_n^p)_n$, whose output Young law on $\rho_{p,g}^{\otimes n}$ is $P_{m_n,p}$ for every n and g .

2.4 Unknown spectrum: randomized rounding

For $N \in \mathbb{N}$, let

$$\mathbb{L}_N = \left\{ x \in \mathbb{Z}^d : \sum_i x_i = N \right\}.$$

For every n such that $m_n > n$, set $\nu_n^{\text{fb}} = (m_n, 0, \dots, 0)$. Given $\mu \in \mathbb{Y}_n$, draw independent variables $U_1, \dots, U_{d-1}$, each uniform on $[-1/2, 1/2)$, and set

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + U_i) + \frac{1}{2} \right\rfloor \quad (1 \leq i < d), \quad \tilde{\nu}_d = m_n - \sum_{i < d} \tilde{\nu}_i. \quad (11)$$

This is a spectrum-independent lattice version of the dilation $\mu \mapsto \gamma_n \mu$: it sends typical labels near np to labels near $m_n p$, with $\tilde{\nu} - \mu$ typically a partition. The uniform shifts randomize the rounding error, avoiding lattice bias in the Gaussian limit. Then $\tilde{\nu} \in \mathbb{L}_{m_n}$. Let $D_n(\tilde{\nu} \mid \mu)$ denote its distribution. Define $K_n^{\text{rnd}} : \mathbb{Y}_n \rightsquigarrow \mathbb{Y}_{m_n}$ by

$$K_n^{\text{rnd}}(\nu \mid \mu) = \begin{cases} D_n(\nu \mid \mu), & \nu \neq \nu_n^{\text{fb}} \text{ and } \nu - \mu \vdash m_n - n, \\ 1 - \sum_{\substack{\lambda \in \mathbb{Y}_{m_n}, \\ \lambda \neq \nu_n^{\text{fb}}, \\ \lambda - \mu \vdash m_n - n}} D_n(\lambda \mid \mu), & \nu = \nu_n^{\text{fb}}, \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

Under (12), every rounded label outside $\mathbb{Y}_{m_n}$, and every $\tilde{\nu} \in \mathbb{Y}_{m_n}$ for which $\tilde{\nu} - \mu \not\vdash m_n - n$, is replaced by ν_n^{fb} , which also retains any mass rounded onto it. Define the amplification branch by

$$\mathcal{C}_n^{\text{univ}} := \mathcal{Q}_n[K_n^{\text{rnd}}].$$

The rounding variables, (12), and the maps $\mathcal{E}_{\mu,\nu}$ are spectrum independent, while $\mathcal{Q}_n[K_n^{\text{rnd}}]$ is $U(d)$ -covariant. Together with the common convention in Section 1.1, this defines the full spectrum-independent, $U(d)$ -covariant sequence $(\mathcal{C}_n^{\text{univ}})_n$.

2.5 Sector and label asymptotics

The following proposition is the only representation-theoretic asymptotic estimate used in the cloning protocols.

Proposition 2.3 (Uniform Cartan-sector fidelity). *Let $K \Subset \Delta_d^\circ$ and let $\delta_n \geq 0$ satisfy $\delta_n \rightarrow 0$. For all sufficiently large n , every $p \in K$, $\mu \in \mathbf{Y}_n$, and $\nu \in \mathbf{Y}_{m_n}$ satisfying*

$$\left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \quad \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n$$

also satisfies $\nu - \mu \vdash m_n - n$. Moreover,

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) = F_{\text{orb}}(\gamma, p) + o(1)$$

uniformly over all such p, μ, ν .

The mechanism is already visible on states obtained by a fixed number of lowering operations. Writing $\omega = \nu - \mu$, the Cartan intertwiner satisfies

$$V_{\mu,\omega} F_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)}) V_{\mu,\omega},$$

where $F_\alpha^{(\lambda)}$ is the lowering operator for $\alpha = e_i - e_j$ on $\mathcal{H}_\lambda$. After normalization, each excitation splits binomially between the two factors, with fraction

$$s_{\alpha,n} = \frac{\mu_i - \mu_j}{\nu_i - \nu_j} = \frac{1}{\gamma} + o(1)$$

entering the μ -factor. Thus the conditional thermal parameter $q_{ij} = p_j/p_i$ is transformed into

$$1 - s_{\alpha,n} + s_{\alpha,n} q_{ij} \longrightarrow 1 - \frac{1 - q_{ij}}{\gamma} =: q_{ij}^{(\gamma)}.$$

The low-weight matrices of the sector output and ideal sector therefore approach those of $\tau_p^{(\gamma)}$ and τ_p , whose fidelity is $F_{\text{orb}}(\gamma, p)$. This is a statement about conditional states for typical label pairs, not convergence of the varying channels $\mathcal{C}_{\mu,\nu}$ to the Gaussian amplifier. The exact PBW calculation and uniform thermal-tail argument that complete the proof are given in Appendix B.

The second estimate concerns only K_n^{rnd} .

Proposition 2.4 (Randomized Young-label dilation). *There is a constant $c_{\text{rnd}} < \infty$, depending only on d and the fixed sequence (γ_n) , such that, for every compact $K \Subset \Delta_d^\circ$ and all sufficiently large n , every $p \in K$ and $\mu \in \mathbf{T}_n(p)$ satisfy*

$$\text{supp } K_n^{\text{rnd}}(\cdot \mid \mu) \subseteq \{\nu \in \mathbf{Y}_{m_n} : \nu - \mu \vdash m_n - n, \quad \|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}\}.$$

For $m_n > n$, let

$$Q_{m_n,p}(\nu) := \sum_{\mu \in \mathbf{Y}_n} P_{n,p}(\mu) K_n^{\text{rnd}}(\nu \mid \mu), \quad \nu \in \mathbf{Y}_{m_n},$$

be the induced output-label law. Then, for every compact $K \Subset \Delta_d^\circ$,

$$F(Q_{m_n,p}, P_{m_n,p}) = F_{\text{cl}}(\gamma, d) + o(1) \quad \text{uniformly for } p \in K. \quad (13)$$

The uniform dither in (11) makes the label kernel a discrete realization of continuous dilation. After centering at Np and scaling by $\sqrt{N}$, embed a label law as a piecewise-constant density on the fundamental cells of the label lattice. Before fallback replacement, the intersection-volume identity (57) identifies the embedded rounded law with the target-cell average of the $\sqrt{\gamma n}$ -dilation of the embedded input law. Write

$$\Sigma_p := (\text{diag}(p) - pp^\top) \Big|_{\{x \in \mathbb{R}^d: \sum_i x_i = 0\}}.$$

The Young local limit is $\mathbf{N}_{0, \Sigma_p}$, so randomized rounding changes the limit to $\mathbf{N}_{0, \gamma \Sigma_p}$, whereas the ideal m_n -label law still converges to $\mathbf{N}_{0, \Sigma_p}$. Their Hellinger affinity is

$$F(\mathbf{N}_{0, \gamma \Sigma_p}, \mathbf{N}_{0, \Sigma_p}) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} = F_{\text{cl}}(\gamma, d).$$

The support check and fallback rule affect only vanishing probability on typical labels.

The proofs of Propositions 2.3 and 2.4 are given in Appendices B and C, respectively.

2.6 Completion of the achievability proofs

Known-spectrum achievability in Theorem 1.1. By the covariance established in Section 2.3, take $g = I$. For K_n^p ,

$$\Pi_{n,p}(\mu, \nu) = P_{n,p}(\mu)P_{m_n,p}(\nu), \quad Q_{m_n,p} = P_{m_n,p}.$$

Let $G_{n,p} = \mathbb{T}_n(p) \times \mathbb{T}_{m_n}(p)$. Young concentration, Lemma 2.1, gives $\Pi_{n,p}(G_{n,p}^c) = o(1)$. Apply Proposition 2.3 with $K = \{p\}$ and

$$\delta_n = n^{-1/3} + m_n^{-1/3}.$$

It follows that every pair in $G_{n,p}$ is compatible for all sufficiently large n , and there is an $\eta_n \rightarrow 0$ such that, uniformly on $G_{n,p}$,

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) \geq F_{\text{orb}}(\gamma, p) - \eta_n.$$

Applying Proposition 2.2 and using $F(Q_{m_n,p}, P_{m_n,p}) = 1$ yields

$$F(\mathcal{C}_n^p(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}) \geq (F_{\text{orb}}(\gamma, p) - \eta_n)(1 - \sqrt{\Pi_{n,p}(G_{n,p}^c)}) = F_{\text{orb}}(\gamma, p) - o(1).$$

□

Proof of part (a) of Theorem 1.2. By the covariance established in Section 2.4, take $g = I$, and fix $K \subseteq \Delta_d^\circ$. Uniformly for $p \in K$, Young concentration, Lemma 2.1, gives $P_{n,p}(\mathbb{T}_n(p)^c) = o(1)$. By Proposition 2.4, for all sufficiently large n , every pair (μ, ν) with $\mu \in \mathbb{T}_n(p)$ and $K_n^{\text{rnd}}(\nu \mid \mu) > 0$ satisfies both $\nu - \mu \vdash m_n - n$ and

$$\left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \frac{\|\nu - \gamma n \mu\|_\infty}{m_n} + \left\| \frac{\mu}{n} - p \right\|_\infty \leq \frac{c_{\text{rnd}}}{m_n} + n^{-1/3}.$$

Apply Proposition 2.3 with

$$\delta_n = n^{-1/3} + \frac{c_{\text{rnd}}}{m_n}.$$

Then, for some $\eta_n \rightarrow 0$, uniformly in p, μ, ν as above,

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) \geq F_{\text{orb}}(\gamma, p) - \eta_n.$$

The joint label law has bad-pair probability $P_{n,p}(\mathbb{T}_n(p)^c)$, and (13) gives $F(Q_{m_n,p}, P_{m_n,p}) = F_{\text{cl}}(\gamma, d) + o(1)$ uniformly on K . Applying Proposition 2.2 therefore gives

$$\begin{aligned} F(\mathcal{C}_n^{\text{univ}}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}) &\geq (F_{\text{orb}}(\gamma, p) - \eta_n) \left(F(Q_{m_n,p}, P_{m_n,p}) - \sqrt{P_{n,p}(\mathbb{T}_n(p)^c)} \right) \\ &= F_{\text{univ}}(\gamma, p) - o(1), \end{aligned}$$

uniformly for $p \in K$. □

3 Optimality from LAN and Gaussian converses

The constructions above establish achievability for the minimax fidelities. We now prove the matching upper bounds by restricting arbitrary cloners to local neighborhoods in which cloning becomes Gaussian dilation and amplification. We first introduce the local coordinates used by the limiting models, then solve the Gaussian problems in Section 3.2, state the two-way LAN approximation in Section 3.3, and finally transfer the resulting converse inequalities back to the finite-dimensional models in Section 3.4.

The LAN transport common to these converse arguments is summarized in Fig. 2. The upper path is the induced Gaussian-model competitor; the lower path is the ideal target against which its fidelity is measured.

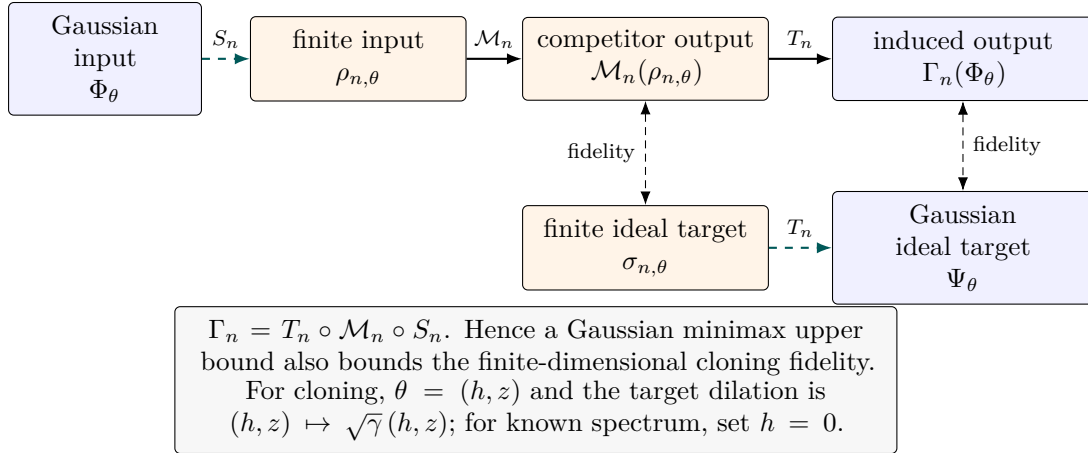


Figure 2: Transporting an arbitrary finite-dimensional cloner through two-way local asymptotic normality. A dashed arrow identifies its displayed endpoint up to uniform $o(1)$ trace-norm error on compact local parameter sets; the solid T_n arrow defines the induced output exactly. The LAN maps are not exact inverses, and the induced channels Γ_n need not be Gaussian or converge. The converse follows from the optimal Gaussian dilation and amplification bounds.

3.1 Local coordinates

Let

$$\mathbb{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad Y(z) = 2 \sum_{j < k} (\bar{z}_{jk} E_{jk} - z_{jk} E_{kj}), \quad z \in \mathbb{C}^{r_d},$$

where E_{jk} are the standard matrix units, and define

$$g_{n,z} = \exp\left(\frac{Y(z)}{\sqrt{n}}\right).$$

In $\rho_{p+h/\sqrt{n}, g_{n,z}}$, the coordinates h and z describe local changes of eigenvalues and eigenbasis, respectively. Since $Y(z)^* = -Y(z)$, one has $g_{n,z} \in \text{U}(d)$. For $L > 0$, set

$$\Omega_L = \{(h, z) \in \mathbb{H} \times \mathbb{C}^{rd} : \|h\|_\infty \leq L, |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

For every fixed $p \in \Delta_d^\circ$ and $L > 0$, the vector $p + h/\sqrt{n}$ belongs to Δ_d° for every $(h, z) \in \Omega_L$ once n is sufficiently large. Under LAN, h becomes a $(d-1)$ -dimensional classical Gaussian shift, while z becomes a collection of displaced oscillator coordinates. The Gaussian theorem below optimizes the two corresponding fidelity losses jointly.

3.2 Gaussian amplification

Let $\mathcal{F}$ be the one-mode Fock space with number basis $\{|k\rangle : k \geq 0\}$, creation and annihilation operators a^*, a , and number operator $\hat{N} = a^*a$. We use the Weyl convention

$$D(z) = \exp(za^* - \bar{z}a). \quad (14)$$

For $0 < q < 1$, set

$$\tau_q = (1-q) \sum_{k \geq 0} q^k |k\rangle\langle k|.$$

Two thermal states commute, and hence

$$F(\tau_q, \tau_{q'}) = \frac{\sqrt{(1-q)(1-q')}}{1 - \sqrt{qq'}}. \quad (15)$$

For $G \geq 1$, put $q^{(G)} = 1 - (1-q)/G$. The quantum-limited phase-insensitive amplifier $\mathcal{A}_G$ satisfies

$$\mathcal{A}_G(D(z)\tau_q D(z)^*) = D(\sqrt{G}z)\tau_{q^{(G)}} D(\sqrt{G}z)^*. \quad (16)$$

Equivalently, $\mathcal{A}_G(X) = \text{Tr}_b[S_r(X \otimes |0\rangle\langle 0|_b)S_r^*]$, where S_r is a two-mode squeezing unitary with $\cosh^2 r = G$. Squeezing supplies the gain $\sqrt{G}$, while tracing out the vacuum idler produces the minimum added noise encoded by $q^{(G)}$.

For $p \in \Delta_d^\circ$, define on $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes rd}$

$$\tau_p = \bigotimes_{i < j} \tau_{q_{ij}}, \quad \tau_p^{(\gamma)} = \bigotimes_{i < j} \tau_{q_{ij}^{(\gamma)}}.$$

With $D(z) = \bigotimes_{i < j} D(z_{ij})$, the three relevant states are

$$\begin{aligned} \text{input:} \quad & \Phi_z^{(p)} = D(z)\tau_p D(z)^*, \\ \text{ideal gain-}\sqrt{\gamma} \text{ target:} \quad & \Psi_z^{(p, \gamma)} = D(\sqrt{\gamma}z)\tau_p D(\sqrt{\gamma}z)^*, \\ \text{amplifier output:} \quad & \mathcal{A}_\gamma^{\otimes rd}(\Phi_z^{(p)}) = D(\sqrt{\gamma}z)\tau_p^{(\gamma)} D(\sqrt{\gamma}z)^*. \end{aligned}$$

Thus the ideal target retains the thermal state τ_p ; only the actual amplifier output contains the additional noise $\tau_p^{(\gamma)}$. By (15),

$$F(\tau_{q^{(\gamma)}}, \tau_q) = f_\gamma(q). \quad (17)$$

Thus, for every $z \in \mathbb{C}^{rd}$, the product-amplifier output and the corresponding ideal state satisfy

$$F(\mathcal{A}_\gamma^{\otimes rd}(\Phi_z^{(p)}), \Psi_z^{(p, \gamma)}) = F_{\text{orb}}(\gamma, p).$$

Equip $\mathbf{H}$ with its induced Euclidean measure and recall

$$\Sigma_p = (\text{diag}(p) - pp^\top) \Big|_{\mathbf{H}} > 0.$$

Write $\mathbf{N}_{h,\Sigma_p}$ for the Gaussian law on $\mathbf{H}$ with mean h and covariance Σ_p . A classical–quantum state has the form

$$R(x) = r(x)\rho_x, \quad r(x) \geq 0, \quad \int_{\mathbf{H}} r(x) \, dx = 1,$$

where x and r are classical, while ρ_x is a quantum state on $\mathcal{F}_{\text{orb}}$. A channel Λ is completely positive and preserves $\int \text{Tr } R(x) \, dx$. Here complete positivity means positivity of $\text{id}_{M_k} \otimes \Lambda$ for every $k \geq 1$, with the matrix factor acting on the quantum trace-class fibers; the classical algebra is abelian. If $S(x) = s(x)\sigma_x$, then

$$F(R, S) = \int_{\mathbf{H}} F(R(x), S(x)) \, dx = \int_{\mathbf{H}} \sqrt{r(x)s(x)} F(\rho_x, \sigma_x) \, dx.$$

With $\|R\|_1 := \int_{\mathbf{H}} \|R(x)\|_1 \, dx$, the usual monotonicity and joint-concavity properties remain valid, and the first equality above is the continuous direct-sum identity for fidelity.

Set

$$\Theta_{h,z}^{\text{in}} = \mathbf{N}_{h,\Sigma_p} \otimes \Phi_z^{(p)}, \quad \Theta_{h,z}^{\text{tar}} = \mathbf{N}_{\sqrt{\gamma}h,\Sigma_p} \otimes \Psi_z^{(p,\gamma)}.$$

Let

$$\mathbf{Z}_L = \{z \in \mathbb{C}^{r_d} : |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

Theorem 3.1 (Gaussian amplification optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\mathbf{Z}_L|} \int_{\mathbf{Z}_L} F\left(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}\right) \, dz = F_{\text{orb}}(\gamma, p), \quad (18)$$

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F\left(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}\right) \, dh \, dz = F_{\text{univ}}(\gamma, p). \quad (19)$$

The first supremum is over all quantum channels on $\mathcal{F}_{\text{orb}}$; the second is over all classical–quantum channels. No Gaussianity, covariance, or product-structure assumption is imposed on the competing channels in either supremum.

The achievability mechanism is explicit. The product amplifier $\mathcal{A}_\gamma^{\otimes r_d}$ attains $F_{\text{orb}}(\gamma, p)$ in the orbital model. In the full model, tensor it with the classical pushforward $x \mapsto \sqrt{\gamma}x$. The resulting Gaussian affinity is $F_{\text{cl}}(\gamma, d)$, so multiplicativity gives $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) = F_{\text{univ}}(\gamma, p)$.

For the converse, average a near-optimal channel over translates of the expanding flat-prior boxes. The vanishing boundary-to-volume error yields a displacement-covariant limiting map. A weighted-fidelity witness then turns the orbital fidelity into a noise moment, which the least-noise inequality bounds by $F_{\text{orb}}(\gamma, p)$. In the full model, covariance forces the classical trace kernel to have convolution form, and the corresponding Gaussian witness contributes at most $F_{\text{cl}}(\gamma, d)$. The two bounds therefore multiply to $F_{\text{univ}}(\gamma, p)$. The compactness, possible lost-mass, and witness arguments are given in [Proposition D.5](#) and [Appendix D](#).

3.3 Local asymptotic normality

Here LAN means that, uniformly for (h, z) in compact sets, the state families $\rho_{p+h/\sqrt{n}, g_{n,z}}^{\otimes n}$ and $\mathbf{N}_{h,\Sigma_p} \otimes \Phi_z^{(p)}$ are asymptotically interconvertible by channels with vanishing trace-norm error. The

Gaussian and oscillator factors represent spectral and eigenbasis fluctuations, respectively; cloning scales (h, z) by $\sqrt{\gamma}$.

The rotation coordinate in [KG09] differs from the oscillator displacement coordinate used here. For $\zeta \in \mathbb{C}^{r_d}$, set

$$X_p(\zeta) = \sum_{j < k} \frac{\zeta_{jk} E_{kj} - \bar{\zeta}_{jk} E_{jk}}{\sqrt{p_j - p_k}}, \quad (L_p \zeta)_{jk} = -\frac{\zeta_{jk}}{2\sqrt{p_j - p_k}}.$$

Then L_p is invertible and

$$X_p(L_p^{-1}z) = Y(z). \quad (20)$$

In the convention of [KG09, Eqs. (4.4) and (4.19)], the physical unitary generated by $X_p(\zeta)$ produces the oscillator state $W(w)^* \tau W(w)$, where W denotes their Weyl operator and $w_{jk} = \zeta_{jk}/(2\sqrt{p_j - p_k})$. Since our convention is $D(z) \tau D(z)^*$ with $D = W$, the corresponding coordinate is $z = -w = L_p \zeta$. Linearity of L_p implies that multiplication of the physical coordinate by $\sqrt{\gamma}$ multiplies z by the same factor.

Theorem 3.2 (Finite-dimensional LAN and target-sample scaling). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{full}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \longrightarrow L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})), \quad S_N^{\text{full}} : L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})) \longrightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, for every compact $C \subset \mathbf{H} \times \mathbb{C}^{r_d}$,

$$\sup_{(h,z) \in C} \left\| T_n^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}}^{\otimes n} \right) - \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right\|_1 \longrightarrow 0, \quad (21)$$

$$\sup_{(h,z) \in C} \left\| S_n^{\text{full}} \left(\mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right) - \rho_{p+h/\sqrt{n}, g_{n,z}}^{\otimes n} \right\|_1 \longrightarrow 0, \quad (22)$$

$$\sup_{(h,z) \in C} \left\| T_{m_n}^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}}^{\otimes m_n} \right) - \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p, \gamma)} \right\|_1 \longrightarrow 0. \quad (23)$$

Proof. The forward and reverse assertions (21)–(22) are [KG09, Theorem 4.3], after identifying their $(d-1)$ -dimensional spectral coordinate with $\mathbf{H}$ and applying the coordinate change (20). For the classical factor, let

$$\iota : \mathbb{R}^{d-1} \longrightarrow \mathbf{H}, \quad \iota(u_1, \dots, u_{d-1}) = \left(u_1, \dots, u_{d-1}, -\sum_{i < d} u_i \right),$$

and let A be the matrix of ι . In the coordinates of [KG09, Eq. (3.3)], the classical covariance is the $(d-1) \times (d-1)$ matrix

$$V(p)_{ij} = \delta_{ij} p_i - p_i p_j, \quad i, j < d.$$

A direct calculation gives

$$AV(p)A^\top = \text{diag}(p) - pp^\top.$$

Hence, when $h = \iota(u)$, the pushforward of their classical Gaussian is $\mathbf{N}_{h, \Sigma_p}$. The coordinate change (20) identifies their oscillator state with $\Phi_z^{(p)}$. Their Schrödinger-picture maps are completely positive and trace preserving on the trace-class and classical–quantum L^1 spaces shown above.

The target-sample assertion (23) is a consequence of the first assertion. Indeed, the identity

$$\rho_{p+h/\sqrt{n}, g_{n,z}} = \rho_{p+\sqrt{\gamma_n}h/\sqrt{m_n}, g_{m_n, \sqrt{\gamma_n}z}}$$

follows from $\sqrt{\gamma_n}/\sqrt{m_n} = 1/\sqrt{n}$ and the real linearity of Y . Apply (21) at sample size m_n on one compact set containing all scaled parameters. Since $\gamma_n \rightarrow \gamma$, translation of Gaussian densities and Weyl conjugation are uniformly trace-norm continuous there, which gives (23). $\square$

Corollary 3.3 (Fixed-spectrum LAN). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{orb}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \rightarrow \mathcal{T}_1(\mathcal{F}_{\text{orb}}), \quad S_N^{\text{orb}} : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \rightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, uniformly for z in compact sets,

$$\|T_n^{\text{orb}}(\rho_{p,g_{n,z}}^{\otimes n}) - \Phi_z^{(p)}\|_1 \rightarrow 0, \quad (24)$$

$$\|S_n^{\text{orb}}(\Phi_z^{(p)}) - \rho_{p,g_{n,z}}^{\otimes n}\|_1 \rightarrow 0, \quad (25)$$

$$\|T_{m_n}^{\text{orb}}(\rho_{p,g_{n,z}}^{\otimes m_n}) - \Psi_z^{(p,\gamma)}\|_1 \rightarrow 0. \quad (26)$$

Proof. Restrict [Theorem 3.2](#) to $h = 0$. The classical Gaussian factor $\mathbf{N}_{0,\Sigma_p}$ is independent of z . Set $T_N^{\text{orb}}(X) := \int_{\mathbf{H}} T_N^{\text{full}}(X)(x) dx$ and $S_N^{\text{orb}}(\sigma) := S_N^{\text{full}}(\mathbf{N}_{0,\Sigma_p} \otimes \sigma)$. The bounds $\|\int_{\mathbf{H}} R(x) dx\|_1 \leq \int_{\mathbf{H}} \|R(x)\|_1 dx$ and $\|\mathbf{N}_{0,\Sigma_p} \otimes X\|_1 = \|X\|_1$ give the three assertions. $\square$

We will use the following continuity estimate here and in the appendices. If A, C, A', C' are positive trace-class operators with trace at most one, then

$$|F(A, C) - F(A', C')| \leq \|A - A'\|_1^{1/2} + \|C - C'\|_1^{1/2}. \quad (27)$$

Indeed, the Powers–Størmer inequality bounds $\|\sqrt{A} - \sqrt{A'}\|_2$ by $\|A - A'\|_1^{1/2}$, and similarly for C, C' . Adding and subtracting $\sqrt{A'}\sqrt{C}$, the reverse triangle inequality and Schatten Hölder give [\(27\)](#), since $\|\sqrt{C}\|_2, \|\sqrt{A'}\|_2 \leq 1$.

The following lemma formalizes this LAN fidelity transfer.

Lemma 3.4 (LAN fidelity transfer). *Let $\{\rho_{n,\theta}\}_{\theta \in C}$ and $\{\sigma_{n,\theta}\}_{\theta \in C}$ be input and target models on a compact parameter set C . Let $\{\Phi_\theta\}$ and $\{\Psi_\theta\}$ be limiting models, and suppose channels S_n, T_n satisfy*

$$\sup_{\theta \in C} \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \rightarrow 0, \quad \sup_{\theta \in C} \|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \rightarrow 0.$$

For every channel $\mathcal{M}_n$, put $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$. Then

$$\inf_{\theta \in C} F(\Gamma_n(\Phi_\theta), \Psi_\theta) \geq \inf_{\theta \in C} F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) - o(1). \quad (28)$$

Proof. Let the two uniform approximation errors be δ_n and η_n . Since $T_n \circ \mathcal{M}_n$ is a channel, uniformly in θ , $\|\Gamma_n(\Phi_\theta) - T_n \mathcal{M}_n(\rho_{n,\theta})\|_1 \leq \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \leq \delta_n$. Also $\|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \leq \eta_n$. Together with [\(27\)](#), these bounds give

$$F(\Gamma_n(\Phi_\theta), \Psi_\theta) \geq F(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) - \sqrt{\delta_n} - \sqrt{\eta_n}.$$

Finally, monotonicity of fidelity under the channel T_n gives

$$F(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) \geq F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}).$$

Taking the infimum over θ proves [\(28\)](#). $\square$

Conditional sector limits versus LAN. Although the same thermal state τ_p appears in both arguments, it represents different limiting operations. The conditional-sector proof fixes the input and output Young labels and, on fixed-height cutoffs, identifies the ideal target with τ_p and the Cartan-sector output with $\tau_p^{(\gamma)}$; see [\(54\)](#) and [\(53\)](#). This comparison directly explains the achievable orbital factor $F_{\text{orb}}(\gamma, p)$. Full LAN instead retains the Young-label fluctuation as $\mathbf{N}_{0,\Sigma_p}$, while orbital LAN marginalizes this classical factor. Thus the common occurrence of τ_p reflects conditioning on Young labels in the sector calculation and marginalizing their fluctuation in orbital LAN. We now combine [Lemma 3.4](#) with [Theorem 3.1](#) to complete the minimax converses.

3.4 Completion of the minimax proofs

We now apply the same transfer principle in three increasingly broad settings. The fixed-spectrum Gaussian bound completes [Theorem 1.1](#); the full Gaussian bound yields the pointwise local minimax lemma below; and that lemma, together with uniform achievability, yields the compact-set statement in [Theorem 1.2](#).

Known-spectrum converse and completion of [Theorem 1.1](#). For the known-spectrum problem, set $u_n(p) := \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$. Choose $\mathcal{M}_n$ with worst-case fidelity at least $u_n(p) - 1/n$. For fixed L , set

$$\Gamma_n = T_{m_n}^{\text{orb}} \circ \mathcal{M}_n \circ S_n^{\text{orb}}.$$

The local orbit $\{g_{n,z} : z \in \mathcal{Z}_L\}$ is contained in the full unitary orbit. Hence [Lemma 3.4](#) gives

$$u_n(p) \leq \inf_{z \in \mathcal{Z}_L} F(\Gamma_n(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}) + o(1).$$

The infimum is no larger than the average. Taking $\limsup_n$, then the supremum over Gaussian-model channels, and finally $L \rightarrow \infty$, [\(18\)](#) gives $\limsup_n u_n(p) \leq F_{\text{orb}}(\gamma, p)$. Together with $\liminf_n u_n(p) \geq F_{\text{orb}}(\gamma, p)$, proved in [Section 2](#), this proves [Theorem 1.1](#). $\square$

The universal penalty cannot be expressed by optimizing over a singleton: if p is fixed in advance, one is back in the known-spectrum problem. The useful pointwise reduction instead considers $n^{-1/2}$ -neighborhoods of a fixed interior spectrum and lets the local window grow after the large- n limit. For all sufficiently large n , set

$$v_{n,L}(p) = \sup_{\mathcal{M}_n} \inf_{(h,z) \in \Omega_L} F(\mathcal{M}_n(\rho_{p+h/\sqrt{n}, g_{n,z}}^{\otimes n}), \rho_{p+h/\sqrt{n}, g_{n,z}}^{\otimes m_n}).$$

Lemma 3.5 (Pointwise local minimax limit). *For every $p \in \Delta_d^\circ$,*

$$\lim_{L \rightarrow \infty} \limsup_{n \rightarrow \infty} v_{n,L}(p) = \lim_{L \rightarrow \infty} \liminf_{n \rightarrow \infty} v_{n,L}(p) = F_{\text{univ}}(\gamma, p).$$

Proof. Fix L . For all sufficiently large n , the spectra $p + h/\sqrt{n}$, $(h, z) \in \Omega_L$, lie in one compact subset of Δ_d° . Applying part (a) of [Theorem 1.2](#) uniformly on this compact set and using continuity of $F_{\text{univ}}(\gamma, \cdot)$ yields

$$\liminf_{n \rightarrow \infty} v_{n,L}(p) \geq F_{\text{univ}}(\gamma, p).$$

For the converse, choose $\mathcal{M}_n$ with worst-case fidelity at least $v_{n,L}(p) - 1/n$, and set

$$\Gamma_n = T_{m_n}^{\text{full}} \circ \mathcal{M}_n \circ S_n^{\text{full}}.$$

By [Lemma 3.4](#),

$$v_{n,L}(p) \leq \inf_{(h,z) \in \Omega_L} F(\Gamma_n(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}) + o(1).$$

Average over Ω_L , take $\limsup_n$, and then use [\(19\)](#) as $L \rightarrow \infty$. This gives

$$\lim_{L \rightarrow \infty} \limsup_{n \rightarrow \infty} v_{n,L}(p) \leq F_{\text{univ}}(\gamma, p).$$

Since both $\limsup_n v_{n,L}(p)$ and $\liminf_n v_{n,L}(p)$ are nonincreasing in L , the two outer limits exist and equal $F_{\text{univ}}(\gamma, p)$. $\square$

Proof of part (b) of Theorem 1.2. For the compact-set problem, set $w_n(K) := \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \mathbf{U}(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$. Part (a) of Theorem 1.2 gives

$$\liminf_{n \rightarrow \infty} w_n(K) \geq \inf_{p \in K} F_{\text{univ}}(\gamma, p).$$

For the reverse inequality, fix $p_0 \in \text{int}_{\mathbf{A}_d} K$. For every fixed L , all local spectra $p_0 + h/\sqrt{n}$, $(h, z) \in \Omega_L$, lie in K for sufficiently large n . Thus $w_n(K) \leq v_{n,L}(p_0)$. Letting first $n \rightarrow \infty$ and then $L \rightarrow \infty$, Lemma 3.5 gives

$$\limsup_{n \rightarrow \infty} w_n(K) \leq F_{\text{univ}}(\gamma, p_0).$$

Take the infimum over $\text{int}_{\mathbf{A}_d} K$. Its closure in $\mathbf{A}_d$ is K , and $F_{\text{univ}}(\gamma, \cdot)$ is continuous, so

$$\limsup_{n \rightarrow \infty} w_n(K) \leq \inf_{p \in K} F_{\text{univ}}(\gamma, p).$$

□

A Strict comparison with purify-and-clone

This appendix compares our cloners with the actual mixed-system output of random purification followed by Werner cloning. We give the details for qubits and then sketch the modewise extension to arbitrary dimension.

Lemma A.1 (Partial trace followed by qubit LAN). *Let $p = (a, b)$, with $a > b > 0$, and set*

$$|\psi_p\rangle = \sqrt{a}|00\rangle + \sqrt{b}|11\rangle, \quad e_s = \sqrt{b}|00\rangle - \sqrt{a}|11\rangle, \quad e_+ = |01\rangle, \quad e_- = |10\rangle.$$

Under the occupation–Fock embedding about ψ_p , partial trace over the second qubit followed by orbital qubit LAN converges strongly on trace-class operators to the one-mode Gaussian channel Λ_p that discards the e_s -mode and whose retained annihilation operator is

$$a_{\text{orb}} = \frac{\sqrt{a}b_- + \sqrt{b}b_+^*}{\sqrt{a-b}}. \quad (29)$$

Here b_+, b_- are the annihilation operators of the two off-diagonal purification-tangent modes.

Proof. For precision, let J_M map the three-mode number vector $|k_s, k_+, k_- \rangle$, $k_s + k_+ + k_- \leq M$, to the corresponding normalized symmetric occupation vector with the remaining particles in ψ_p . If P_M projects onto these number vectors, define the trace-preserving extension $\mathcal{J}_M(X) = J_M P_M X P_M J_M^* + \text{Tr}[(I - P_M)X] |\psi_p\rangle\langle\psi_p|^{\otimes M}$. Let Γ_M be the orbital marginal of the standard full qubit LAN extraction channel [GK06, KG09], with its phase chosen so that the output annihilator corresponds to the positive-CCR fluctuation

$$A_M = \frac{1}{\sqrt{M(a-b)}} \sum_{\ell=1}^M E_{12}^{(\ell)}.$$

Then $\Lambda_M := \Gamma_M \circ \text{Tr}_{E^M} \circ \mathcal{J}_M \rightarrow \Lambda_p$ strongly in trace norm.

It is enough first to test bounded coherent amplitudes $z = (z_s, z_+, z_-)$. For the coherent projector $|z\rangle\langle z|$, the elementary multinomial-to-Poisson limit gives

$$\sup_{\|z\| \leq L} \|\mathcal{J}_M(|z\rangle\langle z|) - |\psi_{z,M}\rangle\langle\psi_{z,M}|^{\otimes M}\|_1 \rightarrow 0$$

for every $L < \infty$, where

$$|\psi_{z,M}\rangle \propto |\psi_p\rangle + M^{-1/2}(z_s e_s + z_+ e_+ + z_- e_-).$$

After partial trace, the product on the right becomes exactly $(\text{Tr}_E |\psi_{z,M}\rangle\langle\psi_{z,M}|)^{\otimes M}$. Smooth spectral decomposition and full qubit LAN, followed by discarding the classical coordinate, therefore give a displaced copy of $\tau_{b/a}$. The first-order off-diagonal entry of the reduced qubit is

$$M^{-1/2}(\sqrt{b}z_+ + \sqrt{a}\bar{z}_-),$$

so, in the canonical orbital normalization, the limiting displacement is $(\sqrt{a}z_- + \sqrt{b}\bar{z}_+)/\sqrt{a-b}$. This is precisely the coherent-state action of Λ_p in (29); the e_s -mode is absent from the retained orbital coordinate.

The relation is canonical because

$$[a_{\text{orb}}, a_{\text{orb}}^*] = \frac{a-b}{a-b} = 1.$$

Together with the complementary mode

$$a_{\perp} = \frac{\sqrt{a} b_+ + \sqrt{b} b_-^*}{\sqrt{a - b}},$$

it is a two-mode squeezing transformation; discarding $a_{\perp}$ gives the claimed Gaussian channel. Finally, the linear span of coherent-state projectors is trace-norm dense in the trace class: a bounded operator annihilating this span has vanishing Husimi Q -symbol, hence is zero by analyticity, and Hahn–Banach gives the density. If Y belongs to this span, contractivity gives

$$\limsup_M \|\Lambda_M(X) - \Lambda_p(X)\|_1 \leq 2\|X - Y\|_1.$$

Approximating X by such Y proves strong convergence; for the thermal input below, the same last step is a number-tail truncation. $\square$

Proposition A.2 (Purify-and-clone for a full-rank qubit). *Let $p = (a, b)$, where $a > b > 0$, put $q = b/a$, and suppose that $m_n/n \rightarrow \gamma > 1$. Denote by $\sigma_{n,m_n}^{\text{RP}}(p)$ the mixed-system output obtained by applying the random-purification channel to $\rho_p^{\otimes n}$, thereby producing n identical copies of a random purification of ρ_p , applying Werner’s pure-state $n \rightarrow m_n$ cloner on $\mathbb{C}^2 \otimes \mathbb{C}^2$, and discarding all purifying registers. Define*

$$q_{\text{RP}} = \frac{\gamma - 1 + \gamma q}{\gamma + (\gamma - 1)q}, \quad R_{\text{RP}}(\gamma, q) = \frac{\sqrt{(1-q)(1-q_{\text{RP}})}}{1 - \sqrt{q q_{\text{RP}}}}. \quad (30)$$

Then

$$\limsup_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{RP}}(p), \rho_p^{\otimes m_n}) \leq R_{\text{RP}}(\gamma, q) < F_{\text{orb}}(\gamma, p). \quad (31)$$

For $\gamma = 2$ and $p = (2/3, 1/3)$, moreover,

$$\begin{aligned} R_{\text{RP}}(2, 1/2) &= \frac{2 + \sqrt{10}}{6} \approx 0.8604 \\ &< F_{\text{univ}}(2, p) \approx 0.8856 < F_{\text{orb}}(2, p) \approx 0.9121. \end{aligned} \quad (32)$$

Thus our known-spectrum cloner is strictly better for every full-rank nondegenerate qubit spectrum, and our spectrum-independent cloner is also strictly better in the displayed example.

Proof. Use the canonical purification ψ_p from [Lemma A.1](#). System-unitary covariance makes the fidelity independent of the chosen representative of the orbit. The random-purification output is the Haar mixture of $((I \otimes U)|\psi_p\rangle\langle\psi_p|(I \otimes U)^*)^{\otimes n}$, with $U \in \text{U}(2)$. Write $\mathcal{W}_{n,m}^{(4)}$ for Werner’s pure-state cloner on $\mathbb{C}^4$. Its unitary covariance and the unitary invariance of the partial trace imply, for every U ,

$$\begin{aligned} &\text{Tr}_{E^m} \mathcal{W}_{n,m}^{(4)}(((I \otimes U)|\psi_p\rangle\langle\psi_p|(I \otimes U)^*)^{\otimes n}) \\ &= \text{Tr}_{E^m} [(I \otimes U)^{\otimes m} \mathcal{W}_{n,m}^{(4)}(|\psi_p\rangle\langle\psi_p|^{\otimes n})(I \otimes U^*)^{\otimes m}] = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(4)}(|\psi_p\rangle\langle\psi_p|^{\otimes n}). \end{aligned}$$

Consequently the Haar average drops out exactly after the purifying registers are discarded:

$$\sigma_{n,m}^{\text{RP}}(p) = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(4)}(|\psi_p\rangle\langle\psi_p|^{\otimes n}). \quad (33)$$

Use the tangent basis e_s, e_+, e_- from [Lemma A.1](#). In the symmetric occupation basis, let $\mathbf{k} = (k_s, k_+, k_-)$ record the numbers of particles in these directions. Werner's symmetric-projection formula shows that its output before the partial trace in (33) is diagonal, with eigenvalue

$$\pi_{n,m}(\mathbf{k}) = \frac{\binom{m-|\mathbf{k}|}{n}}{\binom{m+3}{n+3}}, \quad |\mathbf{k}| \leq m-n, \quad (34)$$

on the occupation vector indexed by $\mathbf{k}$. For fixed $\mathbf{k}$,

$$\pi_{n,m_n}(\mathbf{k}) \longrightarrow \gamma^{-3} \left(\frac{\gamma-1}{\gamma} \right)^{|\mathbf{k}|}.$$

Both sides are normalized probability masses, so the discrete Scheffé lemma gives trace-norm convergence, under the usual occupation–Fock embedding, to three independent thermal modes with parameter $\vartheta = (\gamma-1)/\gamma$, each of mean photon number

$$N = \frac{\vartheta}{1-\vartheta} = \gamma-1. \quad (35)$$

Let ω_n denote this output under the occupation–Fock embedding. Then $\|\omega_n - \tau_{\vartheta}^{\otimes 3}\|_1 \rightarrow 0$, and contractivity together with [Lemma A.1](#) gives

$$\Gamma_{m_n}(\sigma_{n,m_n}^{\text{RP}}(p)) = \Lambda_{m_n}(\omega_n) \longrightarrow \Lambda_p(\tau_{\vartheta}^{\otimes 3})$$

in trace norm. For the tangent vacuum, (29) has mean photon number $b/(a-b)$, and hence gives the ideal mixed-qubit LAN state τ_q . If the two off-diagonal input modes are instead independent thermal states of mean $N = \gamma-1$, their output is thermal with mean

$$N_{\text{RP}} = \frac{aN + b(N+1)}{a-b} = \frac{\gamma-1+b}{a-b}.$$

Its thermal parameter is

$$\frac{N_{\text{RP}}}{N_{\text{RP}}+1} = \frac{\gamma-1+b}{\gamma-1+a} = q_{\text{RP}}.$$

Thus the first limit above is $\tau_{q_{\text{RP}}}$, while ordinary qubit LAN gives $\Gamma_{m_n}(\rho_p^{\otimes m_n}) \rightarrow \tau_q$ in trace norm. Fidelity is nondecreasing under channels and continuous under trace-norm convergence. Therefore, using (15),

$$\limsup_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{RP}}(p), \rho_p^{\otimes m_n}) \leq F(\tau_{q_{\text{RP}}}, \tau_q) = R_{\text{RP}}(\gamma, q).$$

Finally, the optimal mixed-state amplifier has thermal parameter

$$q_{\text{opt}} = \frac{\gamma-1+q}{\gamma}, \quad q_{\text{RP}} - q_{\text{opt}} = \frac{q(\gamma-1)(1-q)}{\gamma[\gamma + (\gamma-1)q]} > 0.$$

For fixed q , set

$$h_q(x) = \frac{\sqrt{(1-q)(1-x)}}{1-\sqrt{qx}}.$$

For $x > q$,

$$\frac{d}{dx} \log h_q(x) = \frac{\sqrt{q} - \sqrt{x}}{2\sqrt{x}(1-x)(1-\sqrt{qx})} < 0.$$

Since $q_{\text{RP}} > q_{\text{opt}} > q$, it follows that

$$R_{\text{RP}}(\gamma, q) = h_q(q_{\text{RP}}) < h_q(q_{\text{opt}}) = F_{\text{orb}}(\gamma, p).$$

For $\gamma = 2$ and $q = 1/2$, one has $q_{\text{RP}} = 4/5$ and $R_{\text{RP}} = (2 + \sqrt{10})/6$; direct substitution into (1) gives the remaining comparisons in (32). $\square$

Higher-dimensional sketch. The same occupation–Fock and partial-trace LAN mechanism is expected to extend to every d ; here we only give the resulting modewise calculation. Take $p_1 > \dots > p_d > 0$, set $q_{ij} = p_j/p_i$, and use the canonical purification $\sum_i \sqrt{p_i} |ii\rangle$. After the Haar average drops out, Werner cloning on $\mathbb{C}^{d^2}$ has the Fock limit $\tau_{\vartheta}^{\otimes(d^2-1)}$, where $\vartheta = (\gamma - 1)/\gamma$, so every purification-tangent mode has mean $N = \gamma - 1$. Identifying a tangent vector with a matrix Z and writing $S = \text{diag}(\sqrt{p_1}, \dots, \sqrt{p_d})$, the differential of partial trace is $Z \mapsto ZS + SZ^*$. Assuming the corresponding general-dimensional partial-trace LAN statement, the modewise calculation gives the classical–quantum LAN output

$$\mathbf{N}_{0, (2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}^{\text{RP}}}, \quad q_{ij}^{\text{RP}} = \frac{\gamma - 1 + \gamma q_{ij}}{\gamma + (\gamma - 1)q_{ij}}.$$

Indeed, the diagonal tangent block adds covariance $2(\gamma - 1)\Sigma_p$, while the two ordered off-diagonal modes for each $i < j$ give the displayed thermal parameter; the complementary purification directions are discarded. The coherent-state and density argument in the lemma applies independently to these blocks.

Writing $\sigma_{n, m_n}^{\text{RP}}(p)$ for the resulting mixed-system output and using h_q from the proof above, full LAN and fidelity monotonicity would yield

$$\begin{aligned} \limsup_{n \rightarrow \infty} F\left(\sigma_{n, m_n}^{\text{RP}}(p), \rho_p^{\otimes m_n}\right) &\leq B_{\text{RP}}(\gamma, p), \\ B_{\text{RP}}(\gamma, p) &:= \left(\frac{\sqrt{2\gamma-1}}{\gamma}\right)^{(d-1)/2} \prod_{i < j} h_{q_{ij}}\left(q_{ij}^{\text{RP}}\right) \\ &< F_{\text{cl}}(\gamma, d) \prod_{i < j} f_{\gamma}(q_{ij}) = F_{\text{univ}}(\gamma, p) < F_{\text{orb}}(\gamma, p). \end{aligned} \tag{36}$$

The strict middle inequality is modewise: the calculation following (31) gives $q_{ij}^{\text{RP}} > (\gamma - 1 + q_{ij})/\gamma$, and $h_{q_{ij}}$ is decreasing there, while $\sqrt{2\gamma-1}/\gamma < 2\sqrt{\gamma}/(1+\gamma)$ for the classical factor. Thus the modewise calculation predicts a strict comparison with both of our cloners in every dimension. This is not a theorem proved here: a complete proof would require the general-dimensional partial-trace LAN statement and its convergence on the relevant thermal inputs.

B Exact-weight PBW calculation of the sector fidelity

This appendix proves Proposition 2.3 by comparing the target sector $\rho_{p, \nu}$ with the Cartan-sector output $\mathcal{C}_{\mu, \nu}(\rho_{p, \mu})$ on finitely many weight spaces near their highest weights, then controlling the remaining target mass. Lemma B.1 shows that normalized lowering-operator vectors become asymptotically orthonormal on these spaces, while Lemma B.2 gives the uniform limit $p^\lambda/s_\lambda(p) \rightarrow \prod_{i < j} (1 - p_j/p_i)$. The Cartan inclusion $V_{\mu, \nu - \mu}$ then splits each lowering operation between $\mathcal{H}_\mu$ and $\mathcal{H}_{\nu - \mu}$ with asymptotically binomial weights, producing the thermal limits τ_p and $\tau_p^{(\gamma)}$ and hence the fidelity $F_{\text{orb}}(\gamma, p)$.

Because the calculations involve only weights of height bounded in terms of L , $O_L(\cdot)$ may depend on L, d, c, C in $cN \leq \Delta_\alpha^\lambda \leq CN$ for $\alpha \in \mathbb{R}_+$, but is uniform in N and all dominant λ satisfying these inequalities.

B.1 PBW vectors on complete weight spaces

Identify the positive roots of $\mathbf{U}(d)$ with

$$\mathbb{R}_+ = \{e_i - e_j : 1 \leq i < j \leq d\}.$$

For $\alpha = e_i - e_j$, set

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij}, \quad H_\alpha = [E_\alpha, F_\alpha] = E_{ii} - E_{jj}, \quad \Delta_\alpha^\lambda = \lambda_i - \lambda_j.$$

Fix a total order on R_+ . For $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{R_+}$, write

$$F^{\mathbf{r}} = \prod_{\alpha \in R_+}^{\prec} F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_{\alpha} r_\alpha.$$

For a nonnegative simple-root combination η , let $\text{ht}(\eta)$ be its simple-root height and put

$$P(\eta) = \left\{ \mathbf{r} : \sum_{\alpha \in R_+} r_\alpha \alpha = \eta \right\}.$$

The set $P(\eta)$ is finite, and the PBW theorem implies that $\{F^{\mathbf{r}} v_\lambda : \mathbf{r} \in P(\eta)\}$ spans $\mathcal{H}_\lambda[\lambda - \eta]$ [FH91]. Define

$$\mathcal{K}_{\lambda, L} = \bigoplus_{\text{ht}(\eta) \leq L} \mathcal{H}_\lambda[\lambda - \eta]. \quad (37)$$

Let $P_{\lambda, L}$ denote the orthogonal projection onto $\mathcal{K}_{\lambda, L}$.

Lemma B.1 (PBW Gram matrices at fixed height). *Let $\lambda = \lambda^{(N)}$ range over dominant weights satisfying*

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in R_+)$$

for fixed $0 < c < C < \infty$. Define

$$\tilde{e}_{\mathbf{r}}^\lambda = \prod_{\alpha \in R_+}^{\prec} \frac{F_\alpha^{r_\alpha}}{\sqrt{r_\alpha!} (\Delta_\alpha^\lambda)^{r_\alpha/2}} v_\lambda. \quad (38)$$

For every fixed L , uniformly over $\text{ht}(\eta) \leq L$ and $\mathbf{r}, \mathbf{s} \in P(\eta)$,

$$\langle \tilde{e}_{\mathbf{r}}^\lambda, \tilde{e}_{\mathbf{s}}^\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} + O_L(N^{-1/2}). \quad (39)$$

Consequently, for all sufficiently large N , the full Gram matrix $G_{\lambda, \eta}$ on every such exact-weight block is invertible. If

$$(e_{\mathbf{r}}^\lambda)_{\mathbf{r} \in P(\eta)} = (\tilde{e}_{\mathbf{r}}^\lambda)_{\mathbf{r} \in P(\eta)} G_{\lambda, \eta}^{-1/2}, \quad (40)$$

then these vectors form an orthonormal basis of $\mathcal{H}_\lambda[\lambda - \eta]$, and

$$e_{\mathbf{r}}^\lambda = \tilde{e}_{\mathbf{r}}^\lambda + O_L(N^{-1/2}) \quad (41)$$

uniformly for $\text{ht}(\eta) \leq L$ and $\mathbf{r} \in P(\eta)$.

Proof. For $\mathbf{r} \in P(\eta)$, every positive root has positive simple-root height, so $|\mathbf{r}| \leq \text{ht}(\eta)$. Put

$$\hat{F}_\alpha = \frac{F_\alpha}{\sqrt{\Delta_\alpha^\lambda}}, \quad \hat{E}_\alpha = \frac{E_\alpha}{\sqrt{\Delta_\alpha^\lambda}}.$$

Each unnormalized Gram entry $\langle v_\lambda, (\hat{F}^{\mathbf{r}})^* \hat{F}^{\mathbf{s}} v_\lambda \rangle$ is therefore a word of length at most $2L$. Commuting root operators cannot increase the word length, and each root has height at most $d - 1$. Hence every intermediate vector lies in

$$\mathcal{K}_{\lambda, R_L}, \quad R_L := 2L(d - 1).$$

For $0 \leq R \leq R_L$, the restrictions

$$\widehat{E}_\alpha : \mathcal{K}_{\lambda,R} \rightarrow \mathcal{K}_{\lambda,R}, \quad \widehat{F}_\alpha : \mathcal{K}_{\lambda,R} \rightarrow \mathcal{K}_{\lambda,R+d-1}$$

satisfy $\|\widehat{E}_\alpha\| + \|\widehat{F}_\alpha\| = O_L(1)$. To see this, decompose $\mathcal{H}_\lambda$ orthogonally into irreducible $\mathfrak{sl}_2(\alpha)$ -summands. If a summand meets the weight space of $\lambda - \eta$, where $\text{ht}(\eta) \leq R$, let t be its level there and M its highest H_α -eigenvalue. The top of this string, of weight $\lambda - \eta + t\alpha$, is again a weight of $\mathcal{H}_\lambda$; hence $\eta - t\alpha$ is a nonnegative simple-root combination and $t \text{ht}(\alpha) \leq \text{ht}(\eta) \leq R$. Moreover,

$$M = \Delta_\alpha^\lambda - \eta(H_\alpha) + 2t = \Delta_\alpha^\lambda + O_L(1).$$

At level t , the F_α - and E_α -coefficients are $\sqrt{(t+1)(M-t)}$ and $\sqrt{t(M-t+1)}$. After division by $\sqrt{\Delta_\alpha^\lambda}$, both are $O_L(1)$, proving the norm bound. All operator norms below are taken after restriction to $\mathcal{K}_{\lambda,R_L}$, with the codomain norm inherited from $\mathcal{H}_\lambda$.

The matrix-unit commutator formula now gives

$$[\widehat{E}_\alpha, \widehat{F}_\beta] = \delta_{\alpha\beta} I + O_L(N^{-1/2}). \quad (42)$$

For $\alpha = \beta$, the left side is $H_\alpha/\Delta_\alpha^\lambda = I + O_L(N^{-1})$ on the cutoff. For $\alpha \neq \beta$, the unnormalized commutator is zero or, up to sign, a root operator E_γ or F_γ . In the nonzero case, the preceding bound gives $\|[\widehat{E}_\alpha, \widehat{F}_\beta]\| = O_L(\sqrt{\Delta_\gamma^\lambda}/\sqrt{\Delta_\alpha^\lambda \Delta_\beta^\lambda}) = O_L(\sqrt{N}/N) = O_L(N^{-1/2})$.

Commute every $\widehat{E}_\alpha$ to the right in $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$. The scalar commutators in (42) contribute only when $\mathbf{r} = \mathbf{s}$, in which case they give $\prod_\alpha r_\alpha!$. Every other term vanishes or contains an $O_L(N^{-1/2})$ remainder. Only $O_L(1)$ terms occur, and all remaining operator factors have norm $O_L(1)$; consequently,

$$\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle = \delta_{\mathbf{r},\mathbf{s}} \prod_\alpha r_\alpha! + O_L(N^{-1/2}).$$

Dividing by the factorials in (38) proves (39).

The number and sizes of the exact-weight blocks are bounded in terms of L , so the entrywise estimate gives $G_{\lambda,\eta} = I + O_L(N^{-1/2})$ in operator norm. Thus $G_{\lambda,\eta}$ is invertible for large N , and the PBW spanning vectors form a basis. Finally, functional calculus gives $G_{\lambda,\eta}^{-1/2} = I + O_L(N^{-1/2})$; hence (40) is an orthonormal basis and satisfies (41). $\square$

B.2 Character normalization and fixed-height matrices

For $a = (a_1, \dots, a_d)$, write $p^a = \prod_i p_i^{a_i}$.

Lemma B.2 (Uniform character normalization). *Let $K \subseteq \Delta_d^\circ$, let $\delta_N \rightarrow 0$, and suppose $p \in K$, $\lambda \vdash N$, and $\|\lambda/N - p\|_\infty \leq \delta_N$. Then, uniformly over this family,*

$$\frac{p^\lambda}{s_\lambda(p)} = \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right) + o(1). \quad (43)$$

Proof. Write the numerator in the Weyl character formula as

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma, \quad A_\sigma = \prod_i p_i^{\lambda_{\sigma(i)} + d - \sigma(i)}.$$

For $\sigma \neq \text{id}$,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} = \sum_i \frac{\lambda_i}{N} \log p_i - \sum_i \frac{\lambda_{\sigma(i)}}{N} \log p_i + O_K(N^{-1}).$$

For $x \in \mathbb{R}^d$, put

$$D_\sigma(x, p) = \sum_i x_i \log p_i - \sum_i x_{\sigma(i)} \log p_i.$$

Strict rearrangement gives $D_\sigma(p, p) > 0$. Since K is compact, the permutations are finite, and the logarithms are uniformly bounded on K , the assumptions $\|\lambda/N - p\|_\infty \leq \delta_N$ and $\delta_N \rightarrow 0$ give $c_K > 0$ such that, for every $\sigma \neq \text{id}$ and all sufficiently large N ,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} \geq c_K, \quad A_\sigma \leq e^{-c_K N} A_{\text{id}}.$$

Consequently,

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma = A_{\text{id}} (1 + O_K(e^{-c_K N})), \quad A_{\text{id}} = p^\lambda \prod_i p_i^{d-i}.$$

Since

$$\det(p_i^{d-j})_{i,j=1}^d = \prod_i p_i^{d-i} \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right),$$

the Weyl character formula gives, uniformly,

$$s_\lambda(p) = p^\lambda \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right)^{-1} (1 + O_K(e^{-c_K N})).$$

This proves (43). $\square$

For the remainder of the appendix, index the number basis of $\mathcal{F}_{\text{orb}}$ by $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbf{R}_+}$, and write it as $\{|\mathbf{r}\rangle\}$. Define the finite-rank projection

$$P_L^{\mathcal{F}} = \sum_{\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L} |\mathbf{r}\rangle \langle \mathbf{r}|. \quad (44)$$

For every fixed L and all sufficiently large N , each highest weight λ satisfying the gap bounds in Lemma B.1 has the orthonormal basis $e_{\mathbf{r}}^\lambda$ of $\mathcal{K}_{\lambda,L}$ constructed there. Define

$$J_{\lambda,L} : \mathcal{K}_{\lambda,L} \longrightarrow P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}, \quad J_{\lambda,L} e_{\mathbf{r}}^\lambda = |\mathbf{r}\rangle. \quad (45)$$

By Lemma B.1, the two spaces have bases indexed by the same $\mathbf{r}$, so $J_{\lambda,L}$ is unitary onto $P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}$. It is defined only on $\mathcal{K}_{\lambda,L}$, and it depends on λ and L .

Proof of Proposition 2.3. Fix K and δ_n as in the proposition, write $m = m_n$, and put $\omega = \nu - \mu$. Since $m/n = \gamma_n \rightarrow \gamma > 1$, we have $m/n - 1 \geq (\gamma - 1)/2$ for all sufficiently large n . Uniformly over the allowed parameters,

$$\begin{aligned} \left\| \frac{\omega}{m-n} - p \right\|_\infty &= \left\| \frac{m/n}{m/n-1} \left(\frac{\nu}{m} - p \right) - \frac{1}{m/n-1} \left(\frac{\mu}{n} - p \right) \right\|_\infty \\ &\leq \frac{m/n+1}{m/n-1} \delta_n = o(1). \end{aligned}$$

Every coordinate and every adjacent gap is uniformly positive on K . Hence $\omega \vdash m - n$ for all sufficiently large n . For $\alpha = e_i - e_j$, set

$$\begin{aligned} s_{\alpha,n} &= \frac{\Delta_\alpha^\mu}{\Delta_\alpha^\nu} = \frac{n}{m} \frac{\mu_i/n - \mu_j/n}{\nu_i/m - \nu_j/m} = \frac{1}{\gamma} + o(1), \\ 1 - s_{\alpha,n} &= \frac{\Delta_\alpha^\omega}{\Delta_\alpha^\nu}. \end{aligned} \quad (46)$$

Consequently, there are constants $0 < c_K < C_K < \infty$, depending only on K and γ , such that, for all large n ,

$$c_K n \leq \Delta_\alpha^\mu, \Delta_\alpha^\nu, \Delta_\alpha^\omega \leq C_K n \quad (\alpha \in \mathbf{R}_+). \quad (47)$$

Let $V = V_{\mu,\omega} : \mathcal{H}_\nu \rightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega$, and write $F_\alpha^{(\lambda)} = d\pi_\lambda(F_\alpha)$ for the action of the root operator on $\mathcal{H}_\lambda$. Intertwining and the highest-weight normalization give

$$VF_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})V.$$

Expanding each $(F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})^{r_\alpha}$ binomially gives the exact identity

$$V\tilde{e}_\mathbf{r}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \tilde{e}_\mathbf{a}^\mu \otimes \tilde{e}_{\mathbf{r}-\mathbf{a}}^\omega, \quad (48)$$

where

$$\beta_{\mathbf{r},\mathbf{a}}^{(n)} = \prod_{\alpha \in \mathbf{R}_+} \sqrt{\binom{r_\alpha}{a_\alpha} s_{\alpha,n}^{a_\alpha} (1 - s_{\alpha,n})^{r_\alpha - a_\alpha}}. \quad (49)$$

If $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$, then (48) has $O_L(1)$ terms, all of height at most L , with $0 \leq \beta_{\mathbf{r},\mathbf{a}}^{(n)} \leq 1$. Since $m/n \rightarrow \gamma > 1$, the sizes $n, m, m - n$ are uniformly comparable, so (47) and Lemma B.1 give $e_\mathbf{b}^\lambda = \tilde{e}_\mathbf{b}^\lambda + O_L(n^{-1/2})$ for $\lambda \in \{\mu, \nu, \omega\}$ and all relevant $\mathbf{b}$. Substituting these relations into (48) and using $\|Vx\| = \|x\|$ gives

$$Ve_\mathbf{r}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} e_\mathbf{a}^\mu \otimes e_{\mathbf{r}-\mathbf{a}}^\omega + O_L(n^{-1/2}) \quad (50)$$

uniformly in norm for $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$.

Put

$$\sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \quad c_\lambda(p) = \frac{p^\lambda}{s_\lambda(p)}, \quad q_\alpha = \frac{p_j}{p_i} \quad (\alpha = e_i - e_j).$$

For $\lambda \in \{\mu, \nu\}$, the weight of $e_\mathbf{b}^\lambda$ is $\lambda - \sum_\alpha b_\alpha \alpha$, and hence

$$\rho_{p,\lambda} e_\mathbf{b}^\lambda = c_\lambda(p) \prod_\alpha q_\alpha^{b_\alpha} e_\mathbf{b}^\lambda.$$

The sector channel gives the local formula

$$\sigma_{\mu,\nu} = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V^*(\rho_{p,\mu} \otimes I_{\mathcal{H}_\omega})V.$$

All asymptotic estimates below are uniform over the allowed family. The character estimate, Lemma B.2, gives

$$c_\lambda(p) = \prod_\alpha (1 - q_\alpha) + o(1) \quad (\lambda = \mu, \nu),$$

while the Weyl dimension formula, (47), and (46) give

$$\frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} = \prod_{\alpha=e_i-e_j} \frac{\Delta_\alpha^\mu + j - i}{\Delta_\alpha^\nu + j - i} = \prod_{\alpha \in \mathbf{R}_+} (s_{\alpha,n} + O(n^{-1})) = \gamma^{-r_d} + o(1).$$

Thus, substituting (50) and the weight action above into this formula, and using $\|\rho_{p,\mu}\| \leq 1$, gives, for fixed L ,

$$\langle e_\mathbf{r}^\nu, \sigma_{\mu,\nu} e_\mathbf{s}^\nu \rangle = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} \sum_{\substack{\mathbf{a} \leq \mathbf{r} \\ \mathbf{b} \leq \mathbf{s}}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \beta_{\mathbf{s},\mathbf{b}}^{(n)} \langle e_\mathbf{a}^\mu, \rho_{p,\mu} e_\mathbf{b}^\mu \rangle \langle e_{\mathbf{r}-\mathbf{a}}^\omega, e_{\mathbf{s}-\mathbf{b}}^\omega \rangle + O_L(n^{-1/2})$$

$$= \delta_{\mathbf{r},\mathbf{s}} \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} c_\mu(p) \prod_{\alpha \in \mathbf{R}_+} (1 - s_{\alpha,n} + s_{\alpha,n} q_\alpha)^{r_\alpha} + O_L(n^{-1/2}), \quad (51)$$

$$\langle e_{\mathbf{r}}^\nu, \rho_{p,\nu} e_{\mathbf{s}}^\nu \rangle = \delta_{\mathbf{r},\mathbf{s}} c_\nu(p) \prod_{\alpha \in \mathbf{R}_+} q_\alpha^{r_\alpha}. \quad (52)$$

Recall the following notation from [Section 3.2](#):

$$q_\alpha^{(\gamma)} = 1 - \frac{1 - q_\alpha}{\gamma}, \quad \tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle \langle k|, \quad \tau_p = \bigotimes_{\alpha \in \mathbf{R}_+} \tau_{q_\alpha}, \quad \tau_p^{(\gamma)} = \bigotimes_{\alpha \in \mathbf{R}_+} \tau_{q_\alpha^{(\gamma)}}.$$

Since $s_{\alpha,n} = \gamma^{-1} + o(1)$, (51) and (52) now give

$$\begin{aligned} \langle e_{\mathbf{r}}^\nu, \sigma_{\mu,\nu} e_{\mathbf{s}}^\nu \rangle &= \delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} (1 - q_\alpha^{(\gamma)}) (q_\alpha^{(\gamma)})^{r_\alpha} + o(1), \\ \langle e_{\mathbf{r}}^\nu, \rho_{p,\nu} e_{\mathbf{s}}^\nu \rangle &= \delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} (1 - q_\alpha) q_\alpha^{r_\alpha} + o(1). \end{aligned}$$

These are precisely the number-basis matrix elements of $\tau_p^{(\gamma)}$ and τ_p . For fixed L , the cutoff space has fixed finite dimension, so the uniform entrywise limits imply convergence in trace norm:

$$\left\| J_{\nu,L} P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0, \quad (53)$$

$$\left\| J_{\nu,L} P_{\nu,L} \rho_{p,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0. \quad (54)$$

Both $\sigma_{\mu,\nu}$ and $\rho_{p,\nu}$ commute with the diagonal torus. The second assertion is immediate; the first follows from torus invariance of $\rho_{p,\mu}$ and covariance of $\mathcal{C}_{\mu,\nu}$. Consequently, $P_{\nu,L}$ reduces both states, and the direct-sum identity gives

$$\begin{aligned} F(\sigma_{\mu,\nu}, \rho_{p,\nu}) &= F(P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) \\ &\quad + F((I - P_{\nu,L}) \sigma_{\mu,\nu} (I - P_{\nu,L}), (I - P_{\nu,L}) \rho_{p,\nu} (I - P_{\nu,L})). \end{aligned}$$

Schatten Hölder gives $F(A, C) \leq \sqrt{\text{Tr } A \text{ Tr } C}$, so the second term is at most

$$\sqrt{\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}]}, \quad (55)$$

because $\text{Tr}[(I - P_{\nu,L}) \sigma_{\mu,\nu}] \leq 1$.

Taking traces in (54) gives, for fixed L ,

$$\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}] = \text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p] + o(1)$$

uniformly as $n \rightarrow \infty$. Since K is compact in Δ_d^0 , there is $b_K < 1$ such that $q_\alpha \leq b_K$ for every $p \in K$ and every positive root α . The occupation numbers under τ_p are independent geometric variables with parameters at most b_K . If their total occupation is s , then their total root height is at most $(d-1)s$. Moreover, the probability of any fixed occupation vector of total occupation s is at most b_K^s , and there are $\binom{s+r_d-1}{r_d-1}$ such vectors. Therefore

$$\sup_{p \in K} \text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p] \leq \sum_{s > L/(d-1)} \binom{s+r_d-1}{r_d-1} b_K^s \longrightarrow 0 \quad (L \rightarrow \infty). \quad (56)$$

The same direct-sum decomposition for $F(\tau_p^{(\gamma)}, \tau_p)$ bounds its complementary part by the square root of the same target thermal tail. Consequently,

$$\begin{aligned} & \left| F(\sigma_{\mu,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \\ & \leq \left| F(P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) - F(P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}}, P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}}) \right| \\ & \quad + \sqrt{\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}]} + \sqrt{\text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p]}. \end{aligned}$$

By (27) and (53)–(54), for every fixed L ,

$$\begin{aligned} & F(P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) \\ & \longrightarrow F(P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}}, P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}}) \end{aligned}$$

uniformly. First let $n \rightarrow \infty$ at fixed L , and then let $L \rightarrow \infty$ using (55)–(56). This proves

$$\left| F(\sigma_{\mu,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \longrightarrow 0$$

uniformly. Finally,

$$F(\tau_p^{(\gamma)}, \tau_p) = \prod_{i < j} F(\tau_{q_{ij}^{(\gamma)}}, \tau_{q_{ij}}) = F_{\text{orb}}(\gamma, p).$$

Here multiplicativity of fidelity follows directly from multiplicativity of the trace norm under tensor products, and the final equality uses (15) and (17). $\square$

C Young local limit and randomized rounding

This appendix proves Proposition 2.4 by expressing the randomized rounding law $D_n(\tilde{\nu} \mid \mu)$ as the normalized intersection volume of the dilated source cell around $\gamma_n \mu$ with the target cell labeled by $\tilde{\nu}$. Lemma C.1 shows that the centered and rescaled target label law $P_{N,p}$ converges uniformly to $\mathbf{N}_{0,\Sigma_p}$, while Lemma C.2 shows that sending $P_{n,p}$ through D_n changes this limit to $\mathbf{N}_{0,\gamma\Sigma_p}$ and yields the label affinity $F_{\text{cl}}(\gamma, d)$. The final proof checks that typical rounded labels satisfy $\nu - \mu \vdash m_n - n$ and that replacement by ν_n^{fb} moves only vanishing mass.

Define the half-open fundamental cell

$$\mathbf{C} = \left\{ \sum_{i=1}^{d-1} t_i (e_i - e_d) : -\frac{1}{2} \leq t_i < \frac{1}{2} \right\} \subset \mathbf{H}.$$

Volumes are taken with the induced $(d-1)$ -dimensional Lebesgue measure. The rounding rule (11) is equivalent, outside a null set of cell boundaries, to choosing the unique $\tilde{\nu} \in \mathbf{L}_{m_n}$ for which

$$\gamma_n(\mu + U) \in \tilde{\nu} + \mathbf{C}, \quad U = \sum_{i=1}^{d-1} U_i (e_i - e_d).$$

For $i < d$, this is precisely the nearest-integer rule because $\lfloor x + \frac{1}{2} \rfloor$ is the unique integer in $(x - \frac{1}{2}, x + \frac{1}{2}]$; the d -th coordinate is then automatic because both sides sum to m_n . Since U is uniform on $\mathbf{C}$, the change of variables $x = \gamma_n(\mu + u)$ gives $dx = \gamma_n^{d-1} du$ and sends the event domain to $\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})$. Consequently,

$$D_n(\tilde{\nu} \mid \mu) = \frac{|\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})|}{\gamma_n^{d-1} |\mathbf{C}|}. \quad (57)$$

All label laws below are regarded as functions on $\mathbf{L}_N$: if a law R is originally defined on $\mathbf{Y}_N$, set $R(\lambda) = 0$ for every $\lambda \in \mathbf{L}_N \setminus \mathbf{Y}_N$.

For $\lambda \in \mathbf{L}_N$, define the recentered and rescaled lattice cell

$$\mathbf{C}_{N,p}(\lambda) = \frac{\lambda - Np + \mathbf{C}}{\sqrt{N}}.$$

For any $R : \mathbf{L}_N \rightarrow \mathbb{R}$ satisfying $\sum_{\lambda \in \mathbf{L}_N} |R(\lambda)| < \infty$, define

$$(\mathcal{E}_{N,p}R)(x) = \frac{N^{(d-1)/2}}{|\mathbf{C}|} R(\lambda) \quad (x \in \mathbf{C}_{N,p}(\lambda)).$$

The cells $\mathbf{C}_{N,p}(\lambda)$, $\lambda \in \mathbf{L}_N$, tile $\mathbf{H}$, and their boundaries have zero induced Lebesgue measure. The map $\mathcal{E}_{N,p}$ preserves ℓ^1 -distance, $\|\mathcal{E}_{N,p}R - \mathcal{E}_{N,p}S\|_1 = \|R - S\|_{\ell^1}$, and, for nonnegative R, S , Hellinger affinity, $\int_{\mathbf{H}} \sqrt{(\mathcal{E}_{N,p}R)(\mathcal{E}_{N,p}S)} dx = \sum_{\lambda \in \mathbf{L}_N} \sqrt{R(\lambda)S(\lambda)}$. Indeed, each $\mathbf{C}_{N,p}(\lambda)$ has volume $|\mathbf{C}|N^{-(d-1)/2}$, which cancels the prefactor in both cellwise integrals.

Lemma C.1 (Uniform Young local limit). *For every compact $K \Subset \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{N,p}P_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (58)$$

Proof. Let $M_{N,p}$ be the multinomial law on nonnegative integer compositions of N , extended by zero to $\mathbf{L}_N$. We first show

$$\sup_{p \in K} \|P_{N,p} - M_{N,p}\|_{\ell^1} \longrightarrow 0. \quad (59)$$

Set $\mathbf{W}_{N,p} = \{\lambda \in \mathbf{L}_N : \|\lambda - Np\|_\infty \leq N^{2/3}\}$. For all large N , $\mathbf{W}_{N,p} \cap \text{supp } M_{N,p} \subset \mathbf{Y}_N$, since $\lambda_i - \lambda_{i+1} \geq N(p_i - p_{i+1}) - 2N^{2/3} > 0$ for $i < d$, uniformly over $p \in K$.

For $\lambda \in \mathbf{Y}_N$, the Weyl dimension and character formulas give

$$P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \dim \mathcal{S}_\lambda = N! \frac{\prod_{i < j} (\lambda_i - \lambda_j + j - i)}{\prod_i (\lambda_i + d - i)!},$$

$$s_\lambda(p) = \frac{\det[p_i^{\lambda_j + d - j}]_{i,j}}{\prod_{i < j} (p_i - p_j)}.$$

Uniformly on $\mathbf{W}_{N,p} \cap \text{supp } M_{N,p}$,

$$s_\lambda(p) = \frac{\prod_i p_i^{\lambda_i + d - i}}{\prod_{i < j} (p_i - p_j)} (1 + O_K(e^{-c_K N}))$$

for some $c_K > 0$. Indeed, both $(\lambda_i + d - i)/N$ and $\log p_i$ are uniformly strictly decreasing there, so strict rearrangement makes every nonidentity determinant term exponentially smaller than the identity term.

Comparison with $M_{N,p}(\lambda) = N! \prod_i p_i^{\lambda_i} / \prod_i \lambda_i!$ gives

$$\frac{P_{N,p}(\lambda)}{M_{N,p}(\lambda)} = (1 + O_K(e^{-c_K N})) \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)} \prod_i \frac{\lambda_i! (Np_i)^{d-i}}{(\lambda_i + d - i)!}.$$

Writing the last product as $\prod_i \prod_{a=1}^{d-i} \frac{Np_i}{\lambda_i + a}$, each factor $\frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)}$, $i < j$, and $\frac{Np_i}{\lambda_i + a}$, $1 \leq a \leq d - i$, equals $1 + O_K(N^{-1/3})$. Therefore

$$\begin{aligned} \|P_{N,p} - M_{N,p}\|_{\ell^1} &\leq \sup_{\lambda \in W_{N,p} \cap \text{supp } M_{N,p}} \left| \frac{P_{N,p}(\lambda)}{M_{N,p}(\lambda)} - 1 \right| \\ &\quad + P_{N,p}(W_{N,p}^c) + M_{N,p}(W_{N,p}^c) \longrightarrow 0 \end{aligned}$$

uniformly for $p \in K$, since the last two terms vanish by [Lemma 2.1](#) and Hoeffding's inequality.

By [\(59\)](#) and the ℓ^1 -isometry of $\mathcal{E}_{N,p}$, it remains to show

$$\sup_{p \in K} \|\mathcal{E}_{N,p} M_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (60)$$

Fix $\beta_0 \in (1/2, 2/3)$ and set

$$V_{N,p} = \{\lambda \in \mathbb{L}_N : \|\lambda - Np\|_\infty \leq N^{\beta_0}\}, \quad x_\lambda = \frac{\lambda - Np}{\sqrt{N}}.$$

Since $\inf_{p \in K} p_d > 0$, $V_{N,p} \subset \text{supp } M_{N,p}$ for all sufficiently large N , uniformly in $p \in K$. Uniformly for $p \in K$ and $\lambda \in V_{N,p}$,

$$\frac{N^{(d-1)/2}}{|\mathbb{C}|} M_{N,p}(\lambda) = \varphi_{0,\Sigma_p}(x_\lambda)(1 + o(1)). \quad (61)$$

To see this, uniform Stirling and third-order Taylor expansions give

$$\log M_{N,p}(\lambda) = -\frac{d-1}{2} \log(2\pi N) - \frac{1}{2} \sum_i \log p_i - \frac{1}{2} \sum_i \frac{x_{\lambda,i}^2}{p_i} + O_K(N^{3\beta_0-2} + N^{\beta_0-1} + N^{-1}).$$

This is the Gaussian exponent since $\langle x, \Sigma_p^{-1} x \rangle = \sum_i x_i^2 / p_i$. The normalizing constants agree since $|\mathbb{C}|^2 = \det[\langle e_i - e_d, e_j - e_d \rangle]_{i,j < d} = d$, while $\ker \Sigma_p = \text{span}\{(1, \dots, 1)\}$ and $\text{cof}_{ii}(\Sigma_p) = \prod_j p_j$ give $\det_{\mathbb{H}} \Sigma_p = d \prod_j p_j$. The remainder is $o(1)$ since $1/2 < \beta_0 < 2/3$.

Set

$$U_{N,p} = \bigcup_{\lambda \in V_{N,p}} \mathbb{C}_{N,p}(\lambda).$$

The multinomial mass outside $V_{N,p}$ and the Gaussian mass outside $U_{N,p}$ are exponentially small: for some $c_K, C_K > 0$,

$$\sup_{p \in K} \left[M_{N,p}(V_{N,p}^c) + \int_{U_{N,p}^c} \varphi_{0,\Sigma_p}(x) dx \right] \leq C_K e^{-c_K N^{2\beta_0-1}}. \quad (62)$$

Indeed, if $\mathbb{C}_{N,p}(\lambda)$ contains x , then $\|x_\lambda\|_\infty \leq \|x\|_\infty + (d-1)/(2\sqrt{N})$. Thus

$$\{x : \|x\|_\infty \leq N^{\beta_0-1/2} - (d-1)/(2\sqrt{N})\} \subset U_{N,p}.$$

On the set at left $\|x_\lambda\|_\infty \leq N^{\beta_0-1/2}$, so $\lambda \in V_{N,p}$. Hoeffding controls $M_{N,p}(V_{N,p}^c)$; the complementary inclusion and the uniform eigenvalue bounds for Σ_p control the Gaussian tail on $U_{N,p}^c$.

The Gaussian density has vanishing relative oscillation on $\mathbb{C}_{N,p}(\lambda)$, $\lambda \in V_{N,p}$. Indeed, uniformly for $p \in K$, $\lambda \in V_{N,p}$, and $x, y, z \in \mathbb{C}_{N,p}(\lambda)$, $\|x - y\| = O(N^{-1/2})$ and $\|z\| = O(N^{\beta_0-1/2})$. Since

$$\nabla \log \varphi_{0,\Sigma_p}(z) = -\Sigma_p^{-1}z,$$

$$\begin{aligned} \left| \log \frac{\varphi_{0,\Sigma_p}(x)}{\varphi_{0,\Sigma_p}(y)} \right| &\leq \sup_{z \in \mathbb{C}_{N,p}(\lambda)} \|\nabla \log \varphi_{0,\Sigma_p}(z)\| \|x - y\| \\ &\leq O_K(N^{\beta_0-1/2}) O(N^{-1/2}) = O_K(N^{\beta_0-1}), \\ \left| \frac{\varphi_{0,\Sigma_p}(x)}{\varphi_{0,\Sigma_p}(y)} - 1 \right| &= O_K(N^{\beta_0-1}) = o(1). \end{aligned} \quad (63)$$

The second estimate follows by exponentiation.

The embedded multinomial density is consequently uniformly accurate on $U_{N,p}$. Indeed, $0 \in \mathbb{C}$, so the point $x_\lambda = (\lambda - Np)/\sqrt{N}$ lies in $\mathbb{C}_{N,p}(\lambda)$. Uniformly for $p \in K$, $\lambda \in V_{N,p}$, and x in the same cell,

$$\begin{aligned} \frac{N^{(d-1)/2} M_{N,p}(\lambda)}{|\mathbb{C}| \varphi_{0,\Sigma_p}(x)} &= \frac{N^{(d-1)/2} M_{N,p}(\lambda)}{|\mathbb{C}| \varphi_{0,\Sigma_p}(x_\lambda)} \frac{\varphi_{0,\Sigma_p}(x_\lambda)}{\varphi_{0,\Sigma_p}(x)} \\ &= (1 + o(1))(1 + o(1)) = 1 + o(1). \end{aligned} \quad (64)$$

The two factors are $1 + o(1)$ by the multinomial local estimate (61) and the Gaussian cell-oscillation bound (63), respectively.

For each $p \in K$,

$$\begin{aligned} \|\mathcal{E}_{N,p} M_{N,p} - \varphi_{0,\Sigma_p}\|_1 &= \int_{U_{N,p}} |(\mathcal{E}_{N,p} M_{N,p})(x) - \varphi_{0,\Sigma_p}(x)| \, dx \\ &\quad + \int_{U_{N,p}^c} |(\mathcal{E}_{N,p} M_{N,p})(x) - \varphi_{0,\Sigma_p}(x)| \, dx \\ &\leq M_{N,p}(V_{N,p}^c) + \int_{U_{N,p}^c} \varphi_{0,\Sigma_p}(x) \, dx \\ &\quad + \sup_{\substack{\lambda \in V_{N,p} \\ x \in \mathbb{C}_{N,p}(\lambda)}} \left| \frac{N^{(d-1)/2} M_{N,p}(\lambda)}{|\mathbb{C}| \varphi_{0,\Sigma_p}(x)} - 1 \right|, \end{aligned}$$

Here the inequality uses $\int_{U_{N,p}} \varphi_{0,\Sigma_p}(x) \, dx \leq 1$ and the cell normalization $\int_{U_{N,p}^c} (\mathcal{E}_{N,p} M_{N,p})(x) \, dx = M_{N,p}(V_{N,p}^c)$. Estimates (62) and (64) make all three terms on the right vanish uniformly, proving (60).

Finally, the ℓ^1 -isometry of $\mathcal{E}_{N,p}$ gives

$$\begin{aligned} \sup_{p \in K} \|\mathcal{E}_{N,p} P_{N,p} - \varphi_{0,\Sigma_p}\|_1 &\leq \sup_{p \in K} \|P_{N,p} - M_{N,p}\|_{\ell^1} \\ &\quad + \sup_{p \in K} \|\mathcal{E}_{N,p} M_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0 \end{aligned}$$

by (59) and (60). □

Lemma C.2 (Continuous limit of randomized dilation). *Let $\tilde{Q}_{m_n,p} = P_{n,p} D_n$ denote the $\tilde{\nu}$ -marginal of the joint law obtained by composing the kernel D_n with $P_{n,p}$, before the replacement in (12). For every compact $K \Subset \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma\Sigma_p}\|_1 \longrightarrow 0, \quad (65)$$

$$\sup_{p \in K} |F(\tilde{Q}_{m_n,p}, P_{m_n,p}) - F_{\text{cl}}(\gamma, d)| \longrightarrow 0. \quad (66)$$

Proof. For $a > 0$ and $f \in L^1(\mathbf{H})$, set

$$(\text{dil}_a f)(y) = a^{-(d-1)/2} f(y/\sqrt{a}),$$

and define the target-cell averaging operator explicitly by

$$(\text{Av}_{m_n,p} f)(x) := \frac{m_n^{(d-1)/2}}{|\mathbf{C}|} \int_{\mathbf{C}_{m_n,p}(\nu)} f(y) \, dy, \quad x \in \mathbf{C}_{m_n,p}(\nu).$$

Since $m_n = \gamma_n n$, the cell definitions and (57) give

$$\begin{aligned} \gamma_n^{-1/2} \mathbf{C}_{m_n,p}(\nu) &= \frac{\gamma_n^{-1}(\nu + \mathbf{C}) - np}{\sqrt{n}}, \\ \left| \gamma_n^{-1/2} \mathbf{C}_{m_n,p}(\nu) \cap \mathbf{C}_{n,p}(\mu) \right| &= \frac{|\mathbf{C}|}{n^{(d-1)/2}} D_n(\nu \mid \mu). \end{aligned}$$

Thus, for $x \in \mathbf{C}_{m_n,p}(\nu)$, changing variables by $z = y/\sqrt{\gamma_n}$ and summing over the source cells gives

$$\begin{aligned} & [\text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p})](x) \\ &= \frac{m_n^{(d-1)/2}}{|\mathbf{C}|} \sum_{\mu \in \mathbf{L}_n} \frac{n^{(d-1)/2}}{|\mathbf{C}|} P_{n,p}(\mu) \left| \gamma_n^{-1/2} \mathbf{C}_{m_n,p}(\nu) \cap \mathbf{C}_{n,p}(\mu) \right| \\ &= \frac{m_n^{(d-1)/2}}{|\mathbf{C}|} \sum_{\mu \in \mathbf{L}_n} P_{n,p}(\mu) D_n(\nu \mid \mu) = (\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p})(x). \end{aligned} \tag{67}$$

For $f, g \in L^1(\mathbf{H})$,

$$\| \text{Av}_{m_n,p} f - \text{Av}_{m_n,p} g \|_1 \leq \| f - g \|_1, \quad \| \text{dil}_{\gamma_n} f - \text{dil}_{\gamma_n} g \|_1 = \| f - g \|_1.$$

By Lemma C.1, $\| \mathcal{E}_{n,p} P_{n,p} - \varphi_{0,\Sigma_p} \|_1 = o(1)$ uniformly on K . Since $\text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} = \varphi_{0,\gamma_n \Sigma_p}$, dilation isometry gives $\| \text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p}) - \varphi_{0,\gamma_n \Sigma_p} \|_1 = o(1)$. Moreover, $\| \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p} \|_1 = o(1)$ uniformly because $\gamma_n \rightarrow \gamma$. Cell averaging satisfies

$$\| \text{Av}_{m_n,p} f - f \|_1 \leq 2^{d-1} \sup_{\substack{h \in \mathbf{H} \\ \|h\| \leq \text{diam}(\mathbf{C})/\sqrt{m_n}}} \| f(\cdot + h) - f \|_1.$$

Since $\gamma_n \rightarrow \gamma$ and K is compact, $\sup_{p \in K, n \geq n_0} \| \nabla \varphi_{0,\gamma_n \Sigma_p} \|_1 < \infty$ for some n_0 . Together with $\| f(\cdot + h) - f \|_1 \leq \| h \| \| \nabla f \|_1$, this gives $\| \text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma_n \Sigma_p} \|_1 = o(1)$. Using the contraction of $\text{Av}_{m_n,p}$ and (67),

$$\begin{aligned} \| \mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma \Sigma_p} \|_1 &= \| \text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p}) - \varphi_{0,\gamma \Sigma_p} \|_1 \\ &\leq \| \text{Av}_{m_n,p} [\text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p}) - \varphi_{0,\gamma_n \Sigma_p}] \|_1 \\ &\quad + \| \text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma_n \Sigma_p} \|_1 + \| \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p} \|_1 \\ &\leq \| \text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p}) - \varphi_{0,\gamma_n \Sigma_p} \|_1 \\ &\quad + \| \text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma_n \Sigma_p} \|_1 + \| \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p} \|_1 = o(1). \end{aligned}$$

The equality is (67). This proves (65).

Applying Lemma C.1 at sample size m_n gives

$$\mathcal{E}_{m_n,p} P_{m_n,p} \longrightarrow \varphi_{0,\Sigma_p}$$

uniformly in L^1 . The embedding preserves Hellinger affinity, and (27) also holds for nonnegative L^1 densities. Therefore

$$F(\tilde{Q}_{m_n,p}, P_{m_n,p}) \longrightarrow F(\mathbf{N}_{0,\gamma\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = F_{\text{cl}}(\gamma, d)$$

uniformly on K . □

Proof of Proposition 2.4. For $i < d$, the nearest-integer rule gives

$$|\tilde{\nu}_i - \gamma_n \mu_i| \leq \frac{\gamma_n + 1}{2}.$$

Since the coordinate sums of $\tilde{\nu}$ and $\gamma_n \mu$ are both m_n ,

$$|\tilde{\nu}_d - \gamma_n \mu_d| \leq \frac{(d-1)(\gamma_n + 1)}{2}.$$

Thus one may take

$$c_{\text{rnd}} = \frac{d-1}{2} \left(1 + \sup_n \gamma_n \right) < \infty.$$

Compactness of K gives $c_0 > 0$ such that, for all large n , every $p \in K$ and $\mu \in \mathbf{T}_n(p)$ satisfy

$$\mu_d \geq c_0 n, \quad \mu_i - \mu_{i+1} \geq c_0 n \quad (i < d).$$

For every $\tilde{\nu} \in \text{supp } D_n(\cdot \mid \mu)$,

$$(\tilde{\nu} - \mu)_d \geq (\gamma_n - 1)\mu_d - c_{\text{rnd}},$$

and

$$(\tilde{\nu}_i - \mu_i) - (\tilde{\nu}_{i+1} - \mu_{i+1}) \geq (\gamma_n - 1)(\mu_i - \mu_{i+1}) - 2c_{\text{rnd}}.$$

These quantities are positive for all large n , uniformly on K . Hence $\tilde{\nu} - \mu$ is a partition; since μ is a partition, so is $\tilde{\nu}$. Moreover, $\tilde{\nu}_d > \mu_d > 0$, so $\tilde{\nu} \neq \nu_n^{\text{fb}}$. Hence, for $\mu \in \mathbf{T}_n(p)$, the law $D_n(\cdot \mid \mu)$ is supported on $\mathbf{Y}_{m_n}$, assigns no mass to ν_n^{fb} , and undergoes no replacement in (12). Consequently,

$$K_n^{\text{rnd}}(\nu \mid \mu) = D_n(\nu \mid \mu), \quad \nu \in \mathbf{Y}_{m_n},$$

and every ν in this support satisfies $\nu - \mu \vdash m_n - n$ and $\|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}$.

Let $\varepsilon_{n,p}$ be the probability, under $\mu \sim P_{n,p}$ and $\tilde{\nu} \sim D_n(\cdot \mid \mu)$, that $\tilde{\nu}$ is replaced in (12). The preceding argument gives

$$\varepsilon_{n,p} \leq P_{n,p}(\mathbf{T}_n(p)^c),$$

which tends uniformly to zero by Lemma 2.1.

By (66), the unreplaced output law has affinity $F_{\text{cl}}(\gamma, d) + o(1)$ with $P_{m_n,p}$, uniformly on K . Moving the rejected mass to ν_n^{fb} changes the law by at most $2\varepsilon_{n,p}$ in ℓ^1 . Applying (27) to classical distributions therefore gives (13). □

D Unified Gaussian flat-prior converse

This appendix proves the two converse inequalities in Theorem 3.1 by averaging any candidate channel Λ over larger parameter boxes and testing the unavoidable noise in its output. Lemma D.1 turns fidelity into such a test expectation. Lemma D.2 shows that any bounded sequence of nearly displacement-covariant maps has a subsequence converging to an exactly covariant map with the same trace bound. Proposition D.3 shows that every covariant map with the required gain is at least as noisy as the amplifier, as measured by the number test, even allowing for lost trace. Lemma D.4

governs the classical-distribution part of a classical–quantum channel: translation covariance forces it to scale x to $\sqrt{\gamma}x$, add independent noise, and possibly lose mass. [Proposition D.5](#) combines these bounds, giving $F_{\text{orb}}(\gamma, p)$ for $\Phi_z^{(p)} \mapsto \Psi_z^{(p, \gamma)}$ and $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p)$ for $\Theta_{h,z}^{\text{in}} \mapsto \Theta_{h,z}^{\text{tar}}$.

The classical parameter space has dimension $k = 0$ for the orbital problem and $k = d - 1$ for the full problem. All trace-class-valued integrals below are Bochner integrals in trace norm.

D.1 Weighted fidelity and compact limits

Lemma D.1 (Weighted root-fidelity bound). *Let R, T be positive trace-class operators, and let W be bounded, positive, and injective. Define*

$$\text{Tr}(TW^{-1}) := \lim_{\varepsilon \downarrow 0} \text{Tr}(T(W + \varepsilon I)^{-1}) \in [0, \infty].$$

If this quantity is finite, then

$$F(R, T)^2 \leq \text{Tr}(RW) \text{Tr}(TW^{-1}). \quad (68)$$

Let $\mathbf{E}$ be a finite-dimensional Euclidean space, let $R \in L^1(\mathbf{E}; \mathcal{T}_1(\mathcal{F}_{\text{orb}}))$ be positive, let g be a probability density on $\mathbf{E}$, and let $\chi > 0$ be measurable. For positive R, S in this L^1 -space, set $F(R, S) = \int_{\mathbf{E}} F(R(y), S(y)) \, dy$, extending the hybrid-state convention in [Section 3](#). The notation gT means the density $y \mapsto g(y)T$. Whenever the right-hand side below is finite,

$$F(R, gT)^2 \leq \left(\int_{\mathbf{E}} \chi(y) \text{Tr}[R(y)W] \, dy \right) \left(\int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right) \text{Tr}(TW^{-1}). \quad (69)$$

Proof. For $W_\varepsilon = W + \varepsilon I$, the Schatten Hölder inequality gives

$$F(R, T) = \|(R^{1/2}W_\varepsilon^{1/2})(W_\varepsilon^{-1/2}T^{1/2})\|_1 \leq \sqrt{\text{Tr}(RW_\varepsilon)} \sqrt{\text{Tr}(TW_\varepsilon^{-1})}.$$

Squaring and letting $\varepsilon \downarrow 0$, using $\text{Tr}(RW_\varepsilon) = \text{Tr}(RW) + \varepsilon \text{Tr} R$ and monotone convergence for the second trace, proves (68). For the hybrid state,

$$F(R, gT) = \int_{\mathbf{E}} \sqrt{g(y)} F(R(y), T) \, dy.$$

Apply the first inequality pointwise and then Cauchy–Schwarz to the factors $\sqrt{\chi(y) \text{Tr}[R(y)W]}$ and $\sqrt{g(y)/\chi(y)}$. This gives (69). $\square$

Lemma D.2 (Compact-observable limits of completely positive maps). *Let $\mathcal{H}_0, \mathcal{H}_1$ be separable Hilbert spaces and let $\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ be completely positive maps with $\sup_j \|\Gamma_j\|_{1 \rightarrow 1} < \infty$. There are a subsequence and a completely positive map $\Gamma : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ such that*

$$\text{Tr}[\Gamma_j(X)K] \longrightarrow \text{Tr}[\Gamma(X)K] \quad (70)$$

for every $X \in \mathcal{T}_1(\mathcal{H}_0)$ and every compact $K \in \mathcal{K}(\mathcal{H}_1)$. If $\text{Tr} \Gamma_j(X) \leq c \text{Tr} X$ for every $X \geq 0$, then $\text{Tr} \Gamma(X) \leq c \text{Tr} X$.

Suppose in addition that $U_g \in \mathcal{U}(\mathcal{H}_0)$ and $V_g \in \mathcal{U}(\mathcal{H}_1)$ are families of unitaries indexed by a set G , and that, for every fixed $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$,

$$\|\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*\|_1 \longrightarrow 0.$$

Then

$$\Gamma(U_g X U_g^*) = V_g \Gamma(X) V_g^*$$

for every $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$.

Proof. Let $M = \sup_j \|\Gamma_j\|_{1 \rightarrow 1}$, and let $\mathcal{D} \subset \mathcal{K}(\mathcal{H}_1)$ be a countable norm-dense $\mathbb{Q}(i)$ -linear $*$ -algebra of finite-rank operators. The predual $\mathcal{T}_1(\mathcal{H}_0)$ is separable, so a diagonal weak- $*$ compactness argument gives a subsequence for which $\Gamma_j^*(K)$ converges weak- $*$ in $\mathcal{B}(\mathcal{H}_0)$ for every $K \in \mathcal{D}$. The bound $\|\Gamma_j^*(K)\| \leq M\|K\|$ extends both the limit and the convergence to every compact K , producing a bounded complex-linear map $\Gamma^* : \mathcal{K}(\mathcal{H}_1) \rightarrow \mathcal{B}(\mathcal{H}_0)$.

To check complete positivity, fix r and $A \geq 0$ in $M_r(\mathcal{K}(\mathcal{H}_1))$. Since $A^{1/2}$ is compact, choose $B_\ell \in M_r(\mathcal{D})$ with $B_\ell \rightarrow A^{1/2}$. Since $\Gamma_j^{*(r)}(B_\ell^* B_\ell) \geq 0$, first letting $j \rightarrow \infty$ and then $\ell \rightarrow \infty$ gives $\Gamma^{*(r)}(A) \geq 0$.

Since $\mathcal{K}(\mathcal{H}_1)^* = \mathcal{T}_1(\mathcal{H}_1)$, define $\Gamma(X)$ by

$$\mathrm{Tr}[\Gamma(X)K] = \mathrm{Tr}[X\Gamma^*(K)] \quad (K \in \mathcal{K}(\mathcal{H}_1)).$$

Then Γ is completely positive and (70) holds. If $P_N \uparrow I$ are finite-rank projections and $X \geq 0$, then

$$\mathrm{Tr} \Gamma(X) = \sup_N \lim_j \mathrm{Tr}[\Gamma_j(X)P_N] \leq c \mathrm{Tr} X.$$

Finally, for fixed g, X and compact K , the operator $V_g^* K V_g$ is compact, and hence

$$\begin{aligned} & \mathrm{Tr}[(\Gamma(U_g X U_g^*) - V_g \Gamma(X) V_g^*) K] \\ &= \lim_j \mathrm{Tr}[(\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*) K] = 0. \end{aligned}$$

Compact operators separate trace-class operators, so the limiting map is covariant. $\square$

D.2 The least-noise oscillator moment

Fix $\gamma > 1$. For $s \geq 1$ and $q_1, \dots, q_s \in (0, 1)$, put

$$q_\ell^{(\gamma)} = 1 - \frac{1 - q_\ell}{\gamma}, \quad \tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell^{(\gamma)}}.$$

The proof below uses the following two external inputs, isolated here to make the dependence on [GM06] explicit.

Fact D.I (Covariant-amplifier representation). *Every displacement-covariant one-mode channel of amplitude gain $\sqrt{\gamma}$ admits an idler realization*

$$a_{\mathrm{out}} = \sqrt{\gamma} a_{\mathrm{in}} + \sqrt{\gamma - 1} b^*.$$

The idler state need not be Gaussian; if the channel is also phase covariant, it may be chosen number diagonal [GM06, Sec. II.C and Appendix; Sec. III for general gain].

Fact D.II (Vacuum-idler stochastic order). *For a thermal input τ_q and a number-diagonal idler, let p_k be the output number distribution of the amplifier in Fact D.I. Then*

$$\sum_{k=0}^m (1 - q^{(\gamma)})(q^{(\gamma)})^k \geq \sum_{k=0}^m p_k \quad (m \geq 0).$$

Thus the vacuum-idler output $\tau_{q^{(\gamma)}}$ is stochastically smallest among these outputs [GM06, Lem. 2.3 and Sec. III].

Proposition D.3 (Least-noise moment). *Let Γ be a completely positive trace-nonincreasing map on the s -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

If

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^s w_{\ell}(\hat{N}_{\ell}),$$

where every w_{ℓ} is bounded, nonnegative, and nonincreasing on $\mathbb{Z}_{\geq 0}$, then

$$\mathrm{Tr}[\Gamma(\tau)W(\hat{\mathbf{N}})] \leq \mathrm{Tr}[\Gamma(\tau)] \mathrm{Tr}[\tau^{(\gamma)}W(\hat{\mathbf{N}})]. \quad (71)$$

Proof. Averaging Γ over independent phase rotations preserves its trace bound and displacement covariance, and it does not change either side of (71). We may therefore assume that Γ is phase covariant in every mode.

Consider one mode. Write $\mathrm{Tr} \Gamma(X) = \mathrm{Tr}(AX)$ with $0 \leq A \leq I$. Displacement covariance implies $D(z)^*AD(z) = A$; hence irreducibility of the Weyl representation gives $A = cI$, where $c = \mathrm{Tr} \Gamma(\tau_q)$. If $c > 0$, then $\Lambda = c^{-1}\Gamma$ is a phase-covariant amplifier of amplitude gain $\sqrt{\gamma}$. By Fact D.I, it has a number-diagonal idler realization. Fact D.II therefore gives

$$\sum_{n=0}^m (1 - q^{(\gamma)})(q^{(\gamma)})^n \geq \sum_{n=0}^m \langle n | \Lambda(\tau_q) | n \rangle \quad (m \geq 0). \quad (72)$$

The distribution on the left is the vacuum-idler output, namely that of $\tau_{q^{(\gamma)}}$, and both output states are number diagonal. For a bounded nonincreasing w , summation by parts writes its expectation as a nonnegative linear combination of these cumulative probabilities. Both distributions in (72) have total mass one, so the common limiting constant cancels. Hence

$$\mathrm{Tr}[\Lambda(\tau_q)w(\hat{N})] \leq \mathrm{Tr}[\tau_{q^{(\gamma)}}w(\hat{N})].$$

Multiplication by c proves the one-mode case; the case $c = 0$ is immediate.

For the multimode case, the claim is immediate if some w_{ℓ} vanishes identically; otherwise, divide each w_{ℓ} by $w_{\ell}(0)$ and assume $0 \leq w_{\ell} \leq 1$. Put

$$W_{\geq \ell} = I_1 \otimes \cdots \otimes I_{\ell-1} \otimes \bigotimes_{j=\ell}^s w_j(\hat{N}_j), \quad W_{\geq s+1} = I,$$

and, for an operator X on mode ℓ , define

$$\Gamma_{\ell}(X) = \mathrm{Tr}_{\neq \ell} \left[W_{\geq \ell+1}^{1/2} \Gamma(\tau_{q_1} \otimes \cdots \otimes \tau_{q_{\ell-1}} \otimes X \otimes \tau_{q_{\ell+1}} \otimes \cdots \otimes \tau_{q_s}) W_{\geq \ell+1}^{1/2} \right].$$

Because $0 \leq W_{\geq \ell+1} \leq I$ and this operator is the identity on mode ℓ , each Γ_{ℓ} is completely positive, trace nonincreasing, and displacement covariant in that mode. The phase-covariant reduction following (71) also makes Γ_{ℓ} phase covariant because the fixed thermal inputs and $W_{\geq \ell+1}$ are phase invariant. Moreover,

$$\mathrm{Tr}[\Gamma_{\ell}(\tau_{q_{\ell}})w_{\ell}(\hat{N}_{\ell})] = \mathrm{Tr}[\Gamma(\tau)W_{\geq \ell}], \quad \mathrm{Tr} \Gamma_{\ell}(\tau_{q_{\ell}}) = \mathrm{Tr}[\Gamma(\tau)W_{\geq \ell+1}].$$

Applying the one-mode inequality to Γ_{ℓ} and iterating over $\ell = 1, \dots, s$ gives (71). $\square$

Specialize to $s = r_d$, $q_\ell = q_{ij}$, $\tau = \tau_p$, and $\tau^{(\gamma)} = \tau_p^{(\gamma)}$. For $\mathbf{k} = (k_{ij})_{i < j} \in \mathbb{Z}_{\geq 0}^{r_d}$, let $t_{\mathbf{k}}$ and $t_{\mathbf{k}}^{(\gamma)}$ be the corresponding number-basis eigenvalues, and define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (73)$$

Equivalently,

$$W_\gamma = \prod_{i < j} w_{ij}(\hat{N}_{ij}), \quad w_{ij}(n) = \sqrt{\frac{1 - q_{ij}}{1 - q_{ij}^{(\gamma)}}} \left(\frac{q_{ij}}{q_{ij}^{(\gamma)}} \right)^{n/2}.$$

Here $\hat{N}_{ij}$ is the number operator of the mode indexed by $i < j$, and $|\mathbf{k}| := \sum_{i < j} k_{ij}$ is the total occupation number. Since $q_{ij} < q_{ij}^{(\gamma)}$, each $w_{ij}(n)$ is positive, bounded, and strictly decreasing in n , and the eigenvalues of W_γ tend to zero as $|\mathbf{k}| \rightarrow \infty$. Thus W_γ is bounded, injective, and compact. Its definition gives

$$\text{Tr}(\tau_p^{(\gamma)} W_\gamma) = \text{Tr}(\tau_p W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = F_{\text{orb}}(\gamma, p), \quad (74)$$

where the inverse is understood as in [Lemma D.1](#).

D.3 Classical trace kernels

Lemma D.4 (Translation-covariant scalar kernels). *Let $\mathbf{E}$ be a finite-dimensional Euclidean space and let $\mathcal{R} : L^1(\mathbf{E}) \rightarrow C_0(\mathbf{E})^*$ be a positive contraction. For a finite Radon measure λ , define $T_a \lambda(A) = \lambda(A - a)$ for Borel A . Suppose*

$$\mathcal{R}(u(\cdot - h)) = T_{\sqrt{\gamma}h} \mathcal{R}(u) \quad (u \in L^1(\mathbf{E}), h \in \mathbf{E}) \quad (75)$$

as an equality of finite Radon measures. Then there is a finite positive measure μ , with $\mu(\mathbf{E}) \leq 1$, such that

$$\mathcal{R}(u)(A) = \int_{\mathbf{E}} u(x) \mu(A - \sqrt{\gamma}x) dx \quad (76)$$

for every $u \in L^1(\mathbf{E})$ and every Borel set A .

Proof. If $\mathbf{E} = \{0\}$, take $\mu = \mathcal{R}(1)$. Otherwise choose a nonnegative approximate identity $u_j \in C_c(\mathbf{E})$ with $\int u_j = 1$ and $\text{supp } u_j \subset \{x : \|x\| \leq j^{-1}\}$. The positive measures $\mathcal{R}(u_j)$ lie in the unit ball of $C_0(\mathbf{E})^*$, so weak-* compactness and separability yield a positive subsequential weak-* limit μ with $\mu(\mathbf{E}) \leq 1$.

For $u \in L^1(\mathbf{E})$, the identity $u * u_j = \int_{\mathbf{E}} u(x) u_j(\cdot - x) dx$ holds as a Bochner integral in $L^1(\mathbf{E})$, so bounded linearity and covariance give

$$\mathcal{R}(u * u_j) = \int_{\mathbf{E}} u(x) T_{\sqrt{\gamma}x} \mathcal{R}(u_j) dx.$$

Pair with $\phi \in C_0(\mathbf{E})$. The approximate-identity property and contractivity give $\mathcal{R}(u * u_j) \rightarrow \mathcal{R}(u)$ in total variation. On the right, weak-* convergence applies pointwise because $y \mapsto \phi(y + \sqrt{\gamma}x)$ lies in $C_0(\mathbf{E})$, and the integrand is bounded by $|u(x)| \|\phi\|_\infty$. Dominated convergence gives

$$\langle \phi, \mathcal{R}(u) \rangle = \int_{\mathbf{E}} u(x) \langle \phi, T_{\sqrt{\gamma}x} \mu \rangle dx.$$

By definition of T_a , the right-hand side pairs ϕ with the finite Radon measure in (76), whose total variation is at most $\|u\|_1 \mu(\mathbf{E})$. Riesz uniqueness proves the claim. $\square$

D.4 One flat-prior converse for both Gaussian models

Proposition D.5 (Flat-prior converse inequality). *Let $k \in \{0, d-1\}$. For $k = 0$, let $\mathbf{E} = \{0\}$, omit the classical factor, and interpret h -integration as evaluation at 0. For $k = d-1$, let $\mathbf{E} = \mathbf{H}$ with its induced Euclidean measure. Let*

$$\mathbf{E}_L = \{h \in \mathbf{E} : \|h\|_\infty \leq L\}, \quad \Xi_L = \mathbf{E}_L \times \mathbf{Z}_L,$$

with $|\mathbf{E}_L| = 1$ when $k = 0$. Define

$$R_{0,z}^{(0)} = \Phi_z^{(p)}, \quad S_{0,z}^{(0)} = \Psi_z^{(p,\gamma)},$$

$$R_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{in}}, \quad S_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{tar}}, \quad c_k = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{k/2}.$$

For $k = 0$, the supremum below is over quantum channels on $\mathcal{F}_{\text{orb}}$; for $k = d-1$, it is over classical-quantum channels. Then

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_{h,z}^{(k)}), S_{h,z}^{(k)}) \, dh \, dz \leq c_k F_{\text{orb}}(\gamma, p). \quad (77)$$

Proof. For $k = 0$, set $f_1 = f_\gamma = \chi = 1$ on the one-point space. For $k = d-1$, let f_1 and f_γ be the centered Gaussian densities with covariances Σ_p and $\gamma\Sigma_p$, and set

$$\chi(y) = \sqrt{\frac{f_1(y)}{f_\gamma(y)}} = \gamma^{k/4} \exp\left[-\frac{\gamma-1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle\right].$$

This function is bounded and strictly positive; for $k = d-1$, it also belongs to $C_0(\mathbf{E})$. In both cases,

$$\int_{\mathbf{E}} \frac{f_1(y)}{\chi(y)} \, dy = \int_{\mathbf{E}} \sqrt{f_1(y)f_\gamma(y)} \, dy = c_k. \quad (78)$$

For $\xi = (h, z)$, define

$$(\mathcal{U}_\xi^{\text{in}} R)(x) = D(z)R(x-h)D(z)^*, \quad (\mathcal{U}_\xi^{\text{out}} R)(y) = D(\sqrt{\gamma}z)R(y-\sqrt{\gamma}h)D(\sqrt{\gamma}z)^*.$$

Then $R_\xi^{(k)} = \mathcal{U}_\xi^{\text{in}}(f_1\tau_p)$ and $S_\xi^{(k)} = \mathcal{U}_\xi^{\text{out}}(f_1\tau_p)$. For a channel Λ , let

$$\bar{\Lambda}_L = \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\eta}^{\text{out}} \circ \Lambda \circ \mathcal{U}_\eta^{\text{in}} \, d\eta.$$

Invariance of F under the isometries $\mathcal{U}_\xi^{\text{out}}$ and joint concavity, applied to the Bochner average, give

$$F_L(\Lambda) = \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_\xi^{(k)}), S_\xi^{(k)}) \, d\xi \leq F(\bar{\Lambda}_L(f_1\tau_p), f_1\tau_p). \quad (79)$$

Let $F_* = \limsup_{L \rightarrow \infty} \sup_{\Lambda} F_L(\Lambda)$. Choose $L_j \rightarrow \infty$ with $\sup_{\Lambda} F_{L_j}(\Lambda) \rightarrow F_*$, and channels $\tilde{\Lambda}_j$ satisfying $F_{L_j}(\tilde{\Lambda}_j) \geq \sup_{\Lambda} F_{L_j}(\Lambda) - j^{-1}$. Set

$$\Lambda_j = \overline{\tilde{\Lambda}_{jL_j}}, \quad R_j = \Lambda_j(f_1\tau_p).$$

Then

$$F_{L_j}(\tilde{\Lambda}_j) \rightarrow F_*, \quad F_{L_j}(\tilde{\Lambda}_j) \leq F(R_j, f_1\tau_p). \quad (80)$$

For $\eta \in \mathbb{E} \times \mathbb{C}^{r_d}$, write

$$A_{j,\eta} = \mathcal{U}_{-\eta}^{\text{out}} \circ \tilde{\Lambda}_j \circ \mathcal{U}_\eta^{\text{in}}, \quad \Lambda_j = \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,\eta} \, d\eta.$$

The group law and the change of variables $u = \eta + \xi$ give

$$\begin{aligned} \Lambda_j(\mathcal{U}_\xi^{\text{in}} R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j} + \xi} A_{j,u}(R) \, du, \\ \mathcal{U}_\xi^{\text{out}} \Lambda_j(R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,u}(R) \, du. \end{aligned}$$

The two integrals cancel on the intersection of the two averaging sets, and $\|A_{j,u}\|_{1 \rightarrow 1} \leq 1$. Therefore, for fixed ξ ,

$$\|\Lambda_j(\mathcal{U}_\xi^{\text{in}} R) - \mathcal{U}_\xi^{\text{out}} \Lambda_j(R)\|_1 \leq \frac{|(\Xi_{L_j} + \xi) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|} \|R\|_1 \longrightarrow 0. \quad (81)$$

Indeed, Ξ_1 is a bounded polytope and $\Xi_L = L\Xi_1$, so the ratio is $O_\xi(L^{-1})$.

Define

$$\Gamma_j^\chi(X) = \int_{\mathbb{E}} \chi(y) [\Lambda_j(f_1 X)](y) \, dy.$$

These maps are completely positive and satisfy $\|\Gamma_j^\chi\|_{1 \rightarrow 1} \leq \|\chi\|_\infty$. Taking $\xi = (0, z)$ in (81), multiplying by χ , and integrating gives

$$\begin{aligned} &\|\Gamma_j^\chi(D(z)XD(z)^*) - D(\sqrt{\gamma}z)\Gamma_j^\chi(X)D(\sqrt{\gamma}z)^*\|_1 \\ &\leq \|\chi\|_\infty \frac{|(\Xi_{L_j} + (0, z)) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|} \|X\|_1 \longrightarrow 0. \end{aligned}$$

Hence [Lemma D.2](#) gives, after passing to a subsequence, a completely positive displacement-covariant limit Γ^χ such that

$$\text{Tr}[\Gamma_j^\chi(X)K] \longrightarrow \text{Tr}[\Gamma^\chi(X)K] \quad (82)$$

for trace-class X and compact K . With

$$M_j = \int_{\mathbb{E}} \chi(y) \text{Tr}[R_j(y)W_\gamma] \, dy = \text{Tr}[\Gamma_j^\chi(\tau_p)W_\gamma],$$

compactness of W_γ gives

$$M_j \longrightarrow \text{Tr}[\Gamma^\chi(\tau_p)W_\gamma]. \quad (83)$$

Moreover, [Lemma D.1](#), (78), and (74) give

$$F(R_j, f_1 \tau_p)^2 \leq M_j c_k F_{\text{orb}}(\gamma, p). \quad (84)$$

If $k = 0$, then $\Gamma_j^\chi = \Lambda_j$ are channels. The trace-bound clause of [Lemma D.2](#) with $c = 1$ makes the limit trace nonincreasing. By [Proposition D.3](#),

$$\begin{aligned} \lim_j M_j &= \text{Tr}[\Gamma^\chi(\tau_p)W_\gamma] \\ &\leq \text{Tr} \Gamma^\chi(\tau_p) \text{Tr}(\tau_p^{(\gamma)} W_\gamma) \leq F_{\text{orb}}(\gamma, p). \end{aligned}$$

Combining (80) with (84) yields $F_* \leq F_{\text{orb}}(\gamma, p)$.

Assume henceforth that $k = d - 1$. For $u \in L^1(\mathbf{E})$, define

$$\mathcal{R}_j(u)(A) = \int_A \text{Tr}([\Lambda_j(u\tau_p)](y)) \, dy.$$

The maps $\mathcal{R}_j : L^1(\mathbf{E}) \rightarrow C_0(\mathbf{E})^*$ are positive contractions. Using countable dense subsets of $L^1(\mathbf{E})$ and $C_0(\mathbf{E})$, pass to a further subsequence converging weak-* on every u to a positive contraction $\mathcal{R}$. Taking $z = 0$ in (81), pairing with a function in $C_0(\mathbf{E})$, and passing to the limit gives

$$\mathcal{R}(u(\cdot - h)) = T_{\sqrt{\gamma}h} \mathcal{R}(u).$$

Let μ be the measure from Lemma D.4; then $\mu(\mathbf{E}) \leq 1$.

Finite-rank projections, (82), the weak-* convergence of $\mathcal{R}_j(f_1)$, and $\chi \in C_0(\mathbf{E})$ give

$$\begin{aligned} \text{Tr } \Gamma^\chi(\tau_p) &\leq \liminf_j \text{Tr } \Gamma_j^\chi(\tau_p) \\ &= \lim_j \int_{\mathbf{E}} \chi(y) \mathcal{R}_j(f_1)(dy) = \int_{\mathbf{E}} \chi(y) \mathcal{R}(f_1)(dy). \end{aligned}$$

By (76), the last integral equals

$$\int_{\mathbf{E}} \mu(dr) \int_{\mathbf{E}} f_1(x) \chi(r + \sqrt{\gamma}x) \, dx.$$

Completing the square gives

$$\int_{\mathbf{E}} f_1(x) \chi(r + \sqrt{\gamma}x) \, dx = c_k \exp\left[-\frac{\gamma - 1}{2\gamma(\gamma + 1)} \langle r, \Sigma_p^{-1}r \rangle\right].$$

Since $\mu(\mathbf{E}) \leq 1$, it follows that $\text{Tr } \Gamma^\chi(\tau_p) \leq c_k$.

Finally write $\text{Tr } \Gamma^\chi(X) = \text{Tr}(AX)$. Covariance implies that A commutes with every Weyl displacement, so $A = cI$, with

$$c = \text{Tr } \Gamma^\chi(\tau_p) \leq c_k \leq 1.$$

Thus Γ^χ is trace nonincreasing, and Proposition D.3 gives

$$\begin{aligned} \lim_j M_j &= \text{Tr}[\Gamma^\chi(\tau_p)W_\gamma] \\ &\leq \text{Tr } \Gamma^\chi(\tau_p) \text{Tr}(\tau_p^{(\gamma)}W_\gamma) \leq c_k F_{\text{orb}}(\gamma, p). \end{aligned}$$

Combining this with (80) and (84) yields

$$F_* \leq c_k F_{\text{orb}}(\gamma, p),$$

which proves the full classical–quantum case. $\square$

References

- [WZ82] W. K. Wootters and W. H. Zurek, A single quantum cannot be cloned, *Nature* **299**, 802–803 (1982).
- [Die82] D. Dieks, Communication by EPR devices, *Physics Letters A* **92**, 271–272 (1982).
- [BH96] V. Buzek and M. Hillery, Quantum copying: Beyond the no-cloning theorem, *Phys. Rev. A* **54**, 1844–1852 (1996).

- [GM97] N. Gisin and S. Massar, Optimal quantum cloning machines, *Phys. Rev. Lett.* **79**, 2153–2156 (1997).
- [BDE+98] D. Bruss, D. P. DiVincenzo, A. Ekert, C. A. Fuchs, C. Macchiavello, and J. A. Smolin, Optimal universal and state-dependent quantum cloning, *Phys. Rev. A* **57**, 2368–2378 (1998), [arXiv:quant-ph/9705038](#).
- [CB99] A. Chefles and S. M. Barnett, Strategies and networks for state-dependent quantum cloning, *Phys. Rev. A* **60**, 136–144 (1999), [arXiv:quant-ph/9812035](#).
- [Wer98] R. F. Werner, Optimal cloning of pure states, *Phys. Rev. A* **58**, 1827–1832 (1998), [arXiv:quant-ph/9804001](#).
- [KW99] M. Keyl and R. F. Werner, Optimal cloning of pure states, testing single clones, *J. Math. Phys.* **40**, 3283–3299 (1999), [arXiv:quant-ph/9807010](#).
- [MP95] S. Massar and S. Popescu, Optimal extraction of information from finite quantum ensembles, *Phys. Rev. Lett.* **74**, 1259–1263 (1995).
- [BEM98] D. Bruss, A. Ekert, and C. Macchiavello, Optimal universal quantum cloning and state estimation, *Phys. Rev. Lett.* **81**, 2598–2601 (1998).
- [BA06] J. Bae and A. Acin, Asymptotic quantum cloning is state estimation, *Phys. Rev. Lett.* **97**, 030402 (2006), [arXiv:quant-ph/0603078](#).
- [CD06] G. Chiribella and G. M. D’Ariano, Quantum information becomes classical when distributed to many users, *Phys. Rev. Lett.* **97**, 250503 (2006).
- [CY14] G. Chiribella and Y. Yang, Optimal asymptotic cloning machines, *New J. Phys.* **16**, 063005 (2014), [arXiv:1404.0990](#).
- [KC22] C. Kim and E. Chitambar, Process-optimized phase-covariant quantum cloning, *Phys. Rev. A* **106**, 022405 (2022), [arXiv:2107.03042](#).
- [CKNWW05] N. J. Cerf, O. Krüger, P. Navez, R. F. Werner, and M. M. Wolf, Non-Gaussian cloning of quantum coherent states is optimal, *Phys. Rev. Lett.* **95**, 070501 (2005), [arXiv:quant-ph/0410058](#).
- [DG98] L.-M. Duan and G.-C. Guo, Probabilistic cloning and identification of linearly independent quantum states, *Phys. Rev. Lett.* **80**, 4999–5002 (1998).
- [NG98] C.-S. Niu and R. B. Griffiths, Optimal copying of one quantum bit, *Phys. Rev. A* **58**, 4377–4393 (1998), [arXiv:quant-ph/9805063](#).
- [BCDM00] D. Bruss, M. Cinchetti, G. M. D’Ariano, and C. Macchiavello, Phase-covariant quantum cloning, *Phys. Rev. A* **62**, 012302 (2000), [arXiv:quant-ph/9909046](#).
- [Cer00] N. J. Cerf, Asymmetric quantum cloning in any dimension, *J. Mod. Opt.* **47**, 187–209 (2000), [arXiv:quant-ph/9805024](#).
- [MJPV99] M. Murao, D. Jonathan, M. B. Plenio, and V. Vedral, Quantum telecloning and multi-particle entanglement, *Phys. Rev. A* **59**, 156–161 (1999), [arXiv:quant-ph/9806082](#).

- [CDP08] G. Chiribella, G. M. D’Ariano, and P. Perinotti, Optimal cloning of unitary transformation, *Phys. Rev. Lett.* **101**, 180504 (2008), [arXiv:0804.0129](#).
- [CYY13] G. Chiribella, Y. Yang, and A. C.-C. Yao, Quantum replication at the Heisenberg limit, *Nat. Commun.* **4**, 2915 (2013), [arXiv:1304.2910](#).
- [GCMBC14] B. Gendra, J. Calsamiglia, R. Muñoz-Tapia, E. Bagan, and G. Chiribella, Probabilistic metrology attains macroscopic cloning of quantum clocks, *Phys. Rev. Lett.* **113**, 260402 (2014), [arXiv:1406.2218](#).
- [DSS15] W. Dür, P. Sekatski, and M. Skotiniotis, Deterministic superreplication of one-parameter unitary transformations, *Phys. Rev. Lett.* **114**, 120503 (2015), [arXiv:1410.6008](#).
- [CYH15] G. Chiribella, Y. Yang, and C. Huang, Universal superreplication of unitary gates, *Phys. Rev. Lett.* **114**, 120504 (2015).
- [CY16] G. Chiribella and Y. Yang, Quantum superreplication of states and gates, *Front. Phys.* **11**, 110304 (2016), [arXiv:1512.05839](#).
- [SSD15] P. Sekatski, M. Skotiniotis, and W. Dür, No-signaling bounds for quantum cloning and metrology, *Phys. Rev. A* **92**, 022355 (2015), [arXiv:1412.3361](#).
- [HT24] Y. Hu and M. Tomamichel, Fundamental limits on quantum cloning from the no-signaling principle, *Phys. Rev. A* **109**, 022221 (2024), [arXiv:2305.02002](#).
- [SGKS25] P. Sekatski, Y. Guryanova, N. B. T. Kothakonda, and M. Skotiniotis, Cloning quantum channels, [arXiv:2509.08059](#) (2025).
- [BGSQ26] V. Brzić, D. Grinko, M. Studziński, and M. T. Quintino, Optimal pure state cloning and transposition are complementary channels, [arXiv:2603.23628](#) (2026).
- [SIGA05] V. Scarani, S. Iblisdir, N. Gisin, and A. Acin, Quantum cloning, *Rev. Mod. Phys.* **77**, 1225–1256 (2005), [arXiv:quant-ph/0511088](#).
- [FWJ+14] H. Fan, Y.-N. Wang, L. Jing, J.-D. Yue, H.-D. Shi, Y.-L. Zhang, and L.-Z. Mu, Quantum cloning machines and the applications, *Phys. Rep.* **544**, 241–322 (2014), [arXiv:1301.2956](#).
- [BCFJS96] H. Barnum, C. M. Caves, C. A. Fuchs, R. Jozsa, and B. Schumacher, Noncommuting mixed states cannot be broadcast, *Phys. Rev. Lett.* **76**, 2818–2821 (1996), [arXiv:quant-ph/9511010](#).
- [KH08] A. Kalev and I. Hen, No-broadcasting theorem and its classical counterpart, *Phys. Rev. Lett.* **100**, 210502 (2008).
- [CEM99] J. I. Cirac, A. K. Ekert, and C. Macchiavello, Optimal purification of single qubits, *Phys. Rev. Lett.* **82**, 4344–4347 (1999), [arXiv:quant-ph/9812075](#).
- [Ras03a] A. E. Rastegin, Upper bound on the global fidelity for mixed-state cloning, *Phys. Rev. A* **67**, 012305 (2003).
- [Ras03b] A. E. Rastegin, A lower bound on the relative error of mixed-state cloning and related operations, *J. Opt. B: Quantum Semiclass. Opt.* **5**, S647–S650 (2003), [arXiv:quant-ph/0208159](#).

- [Ras03c] A. E. Rastegin, Global-fidelity limits of state-dependent cloning of mixed states, *Phys. Rev. A* **68**, 032303 (2003), [arXiv:quant-ph/0301132](#).
- [Fan03] H. Fan, Quantum cloning of mixed states in symmetric subspace, *Phys. Rev. A* **68**, 054301 (2003), [arXiv:quant-ph/0308058](#).
- [FLS07] H. Fan, B. Y. Liu, and K. J. Shi, Quantum cloning of identical mixed qubits, *Quantum Inf. Comput.* **7**, 551–558 (2007), [arXiv:quant-ph/0601017](#).
- [Kaz10] A. Ya. Kazakov, “Partial” quantum cloning and quantum cloning of mixed states, *Int. J. Quantum Inf.* **8**, 435–442 (2010), [arXiv:0711.2860](#).
- [CC07] L. Chen and Y.-X. Chen, Mixed qubit cannot be universally broadcast, *Phys. Rev. A* **75**, 062322 (2007), [arXiv:quant-ph/0612205](#).
- [LW17] M. Lemm and M. M. Wilde, Information-theoretic limitations on approximate quantum cloning and broadcasting, *Phys. Rev. A* **96**, 012304 (2017), [arXiv:1608.07569](#).
- [DMP05] G. M. D’Ariano, C. Macchiavello, and P. Perinotti, Superbroadcasting of mixed states, *Phys. Rev. Lett.* **95**, 060503 (2005), [arXiv:quant-ph/0506251](#).
- [BDMP05] F. Buscemi, G. M. D’Ariano, C. Macchiavello, and P. Perinotti, Optimal superbroadcasting of mixed qubit states, [arXiv:quant-ph/0510155](#) (2005).
- [BDMP06] F. Buscemi, G. M. D’Ariano, C. Macchiavello, and P. Perinotti, Universal and phase-covariant superbroadcasting for mixed qubit states, *Phys. Rev. A* **74**, 042309 (2006), [arXiv:quant-ph/0602125](#).
- [DF07] G.-F. Dang and H. Fan, Optimal broadcasting of mixed states, *Phys. Rev. A* **76**, 022323 (2007), [arXiv:quant-ph/0609182](#).
- [DPS06] G. M. D’Ariano, P. Perinotti, and M. F. Sacchi, Superbroadcasting of continuous variable mixed states, *New J. Phys.* **8**, 99 (2006), [arXiv:quant-ph/0601114](#).
- [PFBC24] A. J. Parzygnat, J. Fullwood, F. Buscemi, and G. Chiribella, Virtual quantum broadcasting, *Phys. Rev. Lett.* **132**, 110203 (2024), [arXiv:2310.13049](#).
- [BGA11] P. Bowles, M. Guță, and G. Adesso, Asymptotically optimal purification and dilution of mixed qubit and Gaussian states, *Phys. Rev. A* **84**, 022320 (2011), [arXiv:1102.2341](#).
- [BGA12] P. Bowles, M. Guță, and G. Adesso, Asymptotically optimal quantum channel reversal for qudit ensembles and multimode Gaussian states, *New J. Phys.* **14**, 113041 (2012), [arXiv:1208.3569](#).
- [Cav82] C. M. Caves, Quantum limits on noise in linear amplifiers, *Phys. Rev. D* **26**, 1817–1839 (1982).
- [CIR00] N. J. Cerf, A. Ipe, and X. Rottenberg, Cloning of continuous quantum variables, *Phys. Rev. Lett.* **85**, 1754–1757 (2000), [arXiv:quant-ph/9909037](#).
- [BCI+01] S. L. Braunstein, N. J. Cerf, S. Iblisdir, P. van Loock, and S. Massar, Optimal cloning of coherent states with a linear amplifier and beam splitters, *Phys. Rev. Lett.* **86**, 4938–4941 (2001), [arXiv:quant-ph/0012046](#).

- [GJ07] M. Guță and A. Jenčová, Local asymptotic normality in quantum statistics, *Commun. Math. Phys.* **276**, 341–379 (2007), [arXiv:quant-ph/0606213](#).
- [GK06] M. Guță and J. Kahn, Local asymptotic normality for qubit states, *Phys. Rev. A* **73**, 052108 (2006), [arXiv:quant-ph/0512075](#).
- [KG09] J. Kahn and M. Guță, Local asymptotic normality for finite dimensional quantum systems, *Commun. Math. Phys.* **289**, 597–652 (2009), [arXiv:0804.3876](#).
- [YFG13] K. Yamagata, A. Fujiwara, and R. D. Gill, Quantum local asymptotic normality based on a new quantum likelihood ratio, *Ann. Statist.* **41**, 2197–2217 (2013), [arXiv:1210.3749](#).
- [GM06] M. Guță and K. Matsumoto, Optimal cloning of mixed Gaussian states, *Phys. Rev. A* **74**, 032305 (2006), [arXiv:quant-ph/0605161](#).
- [ZC17] X. Zhao and G. Chiribella, Quantum amplification and purification of noisy coherent states, *Phys. Rev. A* **95**, 042303 (2017), [arXiv:1701.04602](#).
- [LN24] S. Lahiry and M. Nussbaum, Minimax estimation of low-rank quantum states and their linear functionals, *Bernoulli* **30**, 610–635 (2024), [arXiv:2111.03279](#).
- [KW01] M. Keyl and R. F. Werner, Estimating the spectrum of a density operator, *Phys. Rev. A* **64**, 052311 (2001), [arXiv:quant-ph/0102027](#).
- [PRV67] K. R. Parthasarathy, R. Ranga Rao, and V. S. Varadarajan, Representations of complex semi-simple Lie groups and Lie algebras, *Annals of Mathematics* **85**, 383–429 (1967).
- [FH91] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer (1991).
- [TWZ25] E. Tang, J. Wright, and M. Zhandry, Conjugate queries can help, [arXiv:2510.07622](#) (2025).
- [LTHC26] Z. Li, E. Theil, A. W. Harrow, and I. Chuang, An exponential sample-complexity advantage for coherent quantum inference, [arXiv:2605.21457](#) (2026).
- [LTHC26b] Z. Li, E. Theil, A. W. Harrow, and I. Chuang, Quantum purity amplification for arbitrary eigenstates and multiple outputs, [arXiv:2605.21570](#) (2026).